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United States Patent	12413298
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Ji; Lianghai et al.

Buffer release for inter-satellite link transition

Abstract

Various aspects of the present disclosure generally relate to wireless communication. In some aspects, a wireless communication device may obtain, at an access stratum (AS) layer, an indication of an inter-satellite link (ISL) activation that is to take place at an activation time. The wireless communication device may receive communications. The wireless communication device may buffer the communications in a buffer at the AS layer. The wireless communication device may release the communications from the buffer to an upper protocol layer at a rate of release that is to change between a transition start time and the activation time. Numerous other aspects are described.

Inventors: **Ji; Lianghai (San Diego, CA), Ma; Liangping (San Diego, CA), Ma; Jun (San Diego, CA), Wu; Qiang (San Diego, CA), Sayed Hassan; Mohamad (Paris, FR), Xu; Huilin (Temecula, CA), Duan; Weimin (San Diego, CA), Anantha Swamy; Karthik (La Jolla, CA)**

Applicant: **QUALCOMM Incorporated** (San Diego, CA)

Family ID: **87933631**

Assignee: **QUALCOMM Incorporated (San Diego, CA)**

Appl. No.: **17/932951**

Filed: **September 16, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20240097780 A1	Mar. 21, 2024

Publication Classification

Int. Cl.: **H04B7/185** (20060101)

U.S. Cl.:

CPC **H04B7/18558** (20130101); **H04B7/18563** (20130101); **H04B7/18591** (20130101);

Field of Classification Search

CPC: H04B (7/18558); H04B (7/18563); H04B (7/18591); H04B (7/18541); H04B (7/18521); H04W (36/083); H04W (36/023); H04W (84/005); H04W (88/02)

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Primary Examiner: Rampuria; Sharad

Attorney, Agent or Firm: Harrity & Harrity LLP/QUALCOMM Incorporated

Background/Summary

INTRODUCTION

(1) Aspects of the present disclosure generally relate to wireless communication and to techniques and apparatuses for communications involving an inter-satellite link.

(2) Wireless communication systems are widely deployed to provide various telecommunication services such as telephony, video, data, messaging, and broadcasts. Typical wireless communication systems may employ multiple-access technologies capable of supporting communication with multiple users by sharing available system resources (e.g., bandwidth, transmit power, or the like). Examples of such multiple-access technologies include code division multiple access (CDMA) systems, time division multiple access (TDMA) systems, frequency division multiple access (FDMA) systems, orthogonal frequency division multiple access (OFDMA) systems, single-carrier frequency division multiple access (SC-FDMA) systems, time division synchronous code division multiple access (TD-SCDMA) systems, and Long Term Evolution (LTE). LTE/LTE-Advanced is a set of enhancements to the Universal Mobile Telecommunications System (UMTS) mobile standard promulgated by the Third Generation Partnership Project (3GPP).

(3) A wireless network may include one or more network nodes that support communication for wireless communication devices, such as a user equipment (UE) or multiple UEs. A UE may communicate with a network node via downlink communications and uplink communications. “Downlink” (or “DL”) refers to a communication link from the network node to the UE, and “uplink” (or “UL”) refers to a communication link from the UE to the network node. Some wireless networks may support device-to-device communication, such as via a local link (e.g., a sidelink (SL), a wireless local area network (WLAN) link, and/or a wireless personal area network (WPAN) link, among other examples).

(4) The above multiple access technologies have been adopted in various telecommunication standards to provide a common protocol that enables different UEs to communicate on a municipal, national, regional, and/or global level. New Radio (NR), which may be referred to as 5G, is a set of enhancements to the LTE mobile standard promulgated by the 3GPP. NR is designed to better support mobile broadband internet access by improving spectral efficiency, lowering costs, improving services, making use of new spectrum, and better integrating with other open standards using orthogonal frequency division multiplexing (OFDM) with a cyclic prefix (CP) (CP-OFDM) on the downlink, using CP-OFDM and/or single-carrier frequency division multiplexing (SC-FDM) (also known as discrete Fourier transform spread OFDM (DFT-s-OFDM)) on the uplink, as well as supporting beamforming, multiple-input multiple-output (MIMO) antenna technology, and carrier aggregation. As the demand for mobile broadband access continues to increase, further improvements in LTE, NR, and other radio access technologies remain useful.

SUMMARY

(5) Some aspects described herein relate to a method of wireless communication performed by a wireless communication device. The method may include obtaining, at an access stratum (AS) layer, an indication of an inter-satellite link (ISL) activation that is to take place at an activation time. The method may include receiving communications. The method may include buffering the communications in a buffer at the AS layer. The method may include releasing the communications from the buffer to an upper protocol layer at a rate of release that changes between a transition start time and the activation time.

(6) Some aspects described herein relate to a method of wireless communication performed by a user equipment (UE). The method may include obtaining, at an AS layer, an indication of an ISL deactivation that is to take place at a deactivation time. The method may include buffering

communications in a buffer. The method may include releasing the communications from the buffer at a rate of release that changes between the deactivation time and a transition end time. The method may include transmitting the communications.

(7) Some aspects described herein relate to a wireless communication device for wireless communication. The wireless communication device may include a memory and one or more processors coupled to the memory. The one or more processors may be configured to obtain, at an AS layer, an indication of an ISL activation that is to take place at an activation time. The one or more processors may be configured to receive communications. The one or more processors may be configured to buffer the communications in a buffer at the AS layer. The one or more processors may be configured to release the communications from the buffer to an upper protocol layer at a rate of release that is to change between a transition start time and the activation time.

(8) Some aspects described herein relate to a UE for wireless communication. The UE may include a memory and one or more processors coupled to the memory. The one or more processors may be configured to obtain, at an AS layer, an indication of an ISL deactivation that is to take place at a deactivation time. The one or more processors may be configured to buffer communications in a buffer. The one or more processors may be configured to release the communications from the buffer at a rate of release that is to change between the deactivation time and a transition end time. The one or more processors may be configured to transmit the communications.

(9) Some aspects described herein relate to a non-transitory computer-readable medium that stores a set of instructions for wireless communication by a wireless communication device. The set of instructions, when executed by one or more processors of the wireless communication device, may cause the wireless communication device to obtain, at an AS layer, an indication of an ISL activation that is to take place at an activation time. The set of instructions, when executed by one or more processors of the wireless communication device, may cause the wireless communication device to receive communications. The set of instructions, when executed by one or more processors of the wireless communication device, may cause the wireless communication device to buffer the communications in a buffer at the AS layer. The set of instructions, when executed by one or more processors of the wireless communication device, may cause the wireless communication device to release the communications from the buffer to an upper protocol layer at a rate of release that is to change between a transition start time and the activation time.

(10) Some aspects described herein relate to a non-transitory computer-readable medium that stores a set of instructions for wireless communication by a UE. The set of instructions, when executed by one or more processors of the UE, may cause the UE to obtain, at an AS layer, an indication of an ISL deactivation that is to take place at a deactivation time. The set of instructions, when executed by one or more processors of the UE, may cause the UE to buffer communications in a buffer. The set of instructions, when executed by one or more processors of the UE, may cause the UE to release the communications from the buffer at a rate of release that is to change between the deactivation time and a transition end time. The set of instructions, when executed by one or more processors of the UE, may cause the UE to transmit the communications.

(11) Some aspects described herein relate to an apparatus for wireless communication. The apparatus may include means for obtaining, at an AS layer, an indication of an ISL activation that is to take place at an activation time. The apparatus may include means for receiving communications. The apparatus may include means for buffering the communications in a buffer at the AS layer. The apparatus may include means for releasing the communications from the buffer to an upper protocol layer at a rate of release that changes between a transition start time and the activation time.

(12) Some aspects described herein relate to an apparatus for wireless communication. The apparatus may include means for obtaining, at an AS layer, an indication of an ISL deactivation that is to take place at a deactivation time. The apparatus may include means for buffering communications in a buffer. The apparatus may include means for releasing the communications

from the buffer at a rate of release that changes between the deactivation time and a transition end time. The apparatus may include means for transmitting the communications.
(13) Aspects generally include a method, apparatus, system, computer program product, non-transitory computer-readable medium, UE, base station, network entity, non-terrestrial network (NTN) entity, wireless communication device, and/or processing system as substantially described herein with reference to and as illustrated by the drawings and specification.
(14) The foregoing has outlined rather broadly the features and technical advantages of examples according to the disclosure in order that the detailed description that follows may be better understood. Additional features and advantages will be described hereinafter. The conception and specific examples disclosed may be readily utilized as a basis for modifying or designing other structures for carrying out the same purposes of the present disclosure. Such equivalent constructions do not depart from the scope of the appended claims. Characteristics of the concepts disclosed herein, both their organization and method of operation, together with associated advantages will be better understood from the following description when considered in connection with the accompanying figures. Each of the figures is provided for the purpose of illustration and description, and not as a definition of the limits of the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) So that the above-recited features of the present disclosure can be understood in detail, a more particular description, briefly summarized above, may be had by reference to aspects, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only certain typical aspects of this disclosure and are therefore not to be considered limiting of its scope, for the description may admit to other equally effective aspects. The same reference numbers in different drawings may identify the same or similar elements.
(2) FIG. **1** is a diagram illustrating an example of a wireless network, in accordance with the present disclosure.
(3) FIG. **2** is a diagram illustrating an example of a network entity (e.g., base station) in communication with a user equipment (UE) in a wireless network, in accordance with the present disclosure.
(4) FIG. **3** is a diagram illustrating an example of a disaggregated base station, in accordance with the present disclosure.
(5) FIGS. **4**A-**4**C are diagrams illustrating examples of satellite deployments in a non-terrestrial network, in accordance with the present disclosure.
(6) FIG. **5**A is a diagram illustrating an example of inter-satellite link (ISL) activation, in accordance with the present disclosure.
(7) FIG. **5**B is a diagram illustrating an example ISL deactivation, in accordance with the present disclosure.
(8) FIG. **6** is a diagram illustrating an example of a change in propagation delays associated with an ISL, in accordance with the present disclosure.
(9) FIG. **7** is a diagram illustrating an example of a user plane protocol stack and a control plane protocol stack for a network entity (e.g., base station) and a core network in communication with a UE, in accordance with the present disclosure.
(10) FIG. **8** is a diagram illustrating an example of a gradual increase in latency for received communications, in accordance with the present disclosure.
(11) FIG. **9** is a diagram illustrating an example of a gradual increase in latency for received communications, in accordance with the present disclosure.
(12) FIG. **10** is a diagram illustrating an example process performed, for example, by a wireless

communication device, in accordance with the present disclosure.
(13) FIG. **11** is a diagram illustrating an example process performed, for example, by a UE, in accordance with the present disclosure.
(14) FIG. **12** is a diagram of an example apparatus for wireless communication, in accordance with the present disclosure.
(15) FIG. **13** is a diagram illustrating an example of a hardware implementation for an apparatus employing a processing system, in accordance with the present disclosure.
(16) FIG. **14** is a diagram illustrating an example of an implementation of code and circuitry for an apparatus, in accordance with the present disclosure.
(17) FIG. **15** is a diagram of an example apparatus for wireless communication, in accordance with the present disclosure.
(18) FIG. **16** is a diagram illustrating an example of a hardware implementation for an apparatus employing a processing system, in accordance with the present disclosure.
(19) FIG. **17** is a diagram illustrating an example of an implementation of code and circuitry for an apparatus, in accordance with the present disclosure.

DETAILED DESCRIPTION

(20) A user equipment (UE) may communicate with a network entity (e.g., base station, gateway) using one or more satellites, an unmanned aerial vehicle (UAV), or a high altitude platform station (HAPS). According to one or more examples, a satellite may include a network entity. In some examples, the satellite may be referred to as a non-terrestrial base station, a regenerative repeater, an on-board processing repeater, and/or a non-terrestrial network (NTN) entity. A UE may be stationary or mobile and may be served by a satellite via a service link. The service link may include a link between the satellite and the UE and may include one or more of an uplink or a downlink. The satellite may relay a signal received from a gateway (e.g., ground station) via a feeder link. The feeder link may include a link between the satellite and the gateway and may include one or more of an uplink or a downlink.
(21) An inter-satellite link (ISL), or a link between satellites, may be used to support coverage extension in the NTN (network involving a non-terrestrial entity). For instance, in one example, if a gateway is deployed with certain limitations and a first satellite does not have a direct feeder link connection to the gateway, a second satellite having a feeder link may be communicatively connected to the first satellite with the ISL. In this way, the first satellite has access to the feeder link and service availability improves.
(22) In an example, when the current feeder link of the first satellite deteriorates (e.g., the signal propagation distance increases and/or the radio condition deteriorates), the first satellite may apply an ISL to connect with the second satellite. The second satellite may utilize its feeder link connection and the ISL to facilitate the communication between the first satellite and the network (e.g., the gateway). According to one or more examples, the procedure to switch the first satellite's connection by applying the ISL may be referred to as an “ISL activation” for the first satellite. The ISL activation may help to improve the connectivity between the first satellite and the network. As another example, when the feeder link of the first satellite improves (e.g., the propagation distance decreases and/or the radio condition improves), the first satellite may switch off the ISL and/or disconnect from the second satellite. The first satellite may apply the feeder link to communicate with the network without traffic of the first satellite passing through the second satellite. According to one or more examples, the procedure to disconnect from the second satellite and apply a feeder link connection with the network may be referred to as an “ISL deactivation” for the first satellite. The ISL deactivation may reduce the signal propagation distance and/or the signal propagation trip time between the first satellite and the network.
(23) In some examples, a satellite may be in a geostationary orbit (GSO) or geosynchronous equatorial orbit (GEO), which may be, for example, 36,000 kilometers (km) above the Earth. The speed of the satellite with respect to Earth may be negligible but may have a round-trip propagation

delay of more than 500 milliseconds (ms), as compared to 25 ms for a low Earth orbit (LEO) satellite at 600 km above the Earth or 100 ms for a medium Earth orbit (MEO) satellite. In an NTN, where a distance between UE **120** and a satellite can be larger than 600 km changes in a path loss of a signal may not directly correspond to changes in a propagation distance of the signal. In addition, a satellite may not be always available to the UE **120** if the satellite is in a non-geostationary orbit (NGSO) because the satellite may move where there is no line of sight.

(24) In another example, the first satellite may have a feeder link, but a topology change event may occur where there is a change in a communication path in the NTN topology (how communication links are arranged among communicating entities). This may include a change in which communication link is used or a change in an operation of an NTN entity (e.g., activation, deactivation, no line of sight, out of range). In an example, the topology change event may be an ISL activation, where the first satellite no longer uses the feeder link between the first satellite and the gateway and switches to use of the ISL between the first satellite and the second satellite, which in turn may use a respective feeder link to the gateway. A topology change event may also include an ISL deactivation, where the first satellite switches from using the ISL to using the feeder link.

(25) According to one or more examples, when an ISL is activated or deactivated, the signal propagation delay between the UE and the gateway changes because the time for signal propagation through one satellite is shorter than the time for signal propagation through two satellites separated by a significant distance. Accordingly, the communication service will experience a sudden change in latency. The amount of latency change is related to the propagation delay change. The sudden change in latency, if large, can impact the UE's communication service because the UE's communication service may not be aware of the reason for such a sudden change. Even smaller changes in latency can affect the UE's communication service if sudden. The sudden change in latency may cause unexpected communication service behavior, such as a communication service timeout, triggering of a congestion control algorithm, or change in the quality of the communication service experience.

(26) The UE may obtain, at an access stratum (AS) layer (e.g., physical protocol layer), an indication of an ISL activation that is to take place at an activation time or an ISL deactivation that is to take place at a deactivation time. According to various aspects described herein, instead of an immediate change in the propagation delay, a UE may gradually increase or decrease the latency between a first propagation delay before a topology change event (e.g., satellite switch, activation of ISL, deactivation of ISL) and/or a second propagation delay after the topology change event. This may include changing a rate of release of communications from a buffer of the UE so that the change in the propagation delay experienced by a service application of the UE transitions gradually from the first propagation delay to the second propagation delay, rather than changes instantly. The buffer may be storage that buffers communications (stores for later release or temporarily stores). For the UE receiving communications in association with an ISL activation, the UE may buffer received communications and release the communications from the buffer at the AS layer to an upper protocol layer (e.g., layer above the AS layer, application layer for an application providing a service) at a rate of release that changes between a transition start time and the ISL activation time. The transition start time may be an estimated time to start changing the rate of release. For the UE transmitting communications in association with an ISL deactivation, the UE may release the communications from the buffer at a rate of release that changes between an ISL deactivation time and a transition end time. The transition end time may be an estimated time to stop changing the rate of release. For example, the rate of release may change based at least in part on a configured function (e.g., linear function, stepped function, non-linear function). In this way, the UE may avoid sudden changes in latency and thus avoid communication timeouts, the triggering of traffic congestion algorithms, unexpected changes in the communication experience, or other results that may drop service due to sudden changes in latency or consume additional resources when reconnecting the service. As a result, the UE may reduce latency and conserve

processing resources and signaling resources.

(27) Various aspects of the disclosure are described more fully hereinafter with reference to the accompanying drawings. This disclosure may, however, be embodied in many different forms and should not be construed as limited to any specific structure or function presented throughout this disclosure. Rather, these aspects are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the disclosure to those skilled in the art. One skilled in the art should appreciate that the scope of the disclosure is intended to cover any aspect of the disclosure disclosed herein, whether implemented independently of or combined with any other aspect of the disclosure. For example, an apparatus may be implemented or a method may be practiced using any number of the aspects set forth herein. In addition, the scope of the disclosure is intended to cover such an apparatus or method which is practiced using other structure, functionality, or structure and functionality in addition to or other than the various aspects of the disclosure set forth herein. It should be understood that any aspect of the disclosure disclosed herein may be embodied by one or more elements of a claim.

(28) Several aspects of telecommunication systems will now be presented with reference to various apparatuses and techniques. These apparatuses and techniques will be described in the following detailed description and illustrated in the accompanying drawings by various blocks, modules, components, circuits, steps, processes, algorithms, or the like (collectively referred to as "elements"). These elements may be implemented using hardware, software, or combinations thereof. Whether such elements are implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system.

(29) While aspects may be described herein using terminology commonly associated with a 5G or New Radio (NR) radio access technology (RAT), aspects of the present disclosure can be applied to other RATs, such as a 3G RAT, a 4G RAT, and/or a RAT subsequent to 5G (e.g., 6G).

(30) FIG. **1** is a diagram illustrating an example of a wireless network **100**, in accordance with the present disclosure. The wireless network **100** may be or may include elements of a 5G (e.g., NR) network and/or a 4G (e.g., Long Term Evolution (LTE)) network, among other examples. The wireless network **100** may include a UE **120** or multiple UEs **120** (shown as a UE **120***a*, a UE **120***b*, a UE **120***c*, a UE **120***d*, and a UE **120***e*). The wireless network **100** may also include one or more network entities, such as base stations **110** (shown as a BS **110***a*, a BS **110***b*, a BS **110***c*, and a BS **110***d*), and/or other network entities. A base station **110** is a network entity that communicates with UEs **120**. A base station **110** (sometimes referred to as a BS) may include, for example, an NR base station, an LTE base station, a Node B, an eNB (e.g., in 4G), a gNB (e.g., in 5G), an access point, and/or a transmission reception point (TRP). Each base station **110** may provide communication coverage for a particular geographic area. In the Third Generation Partnership Project (3GPP), the term "cell" can refer to a coverage area of a base station **110** and/or a base station subsystem serving this coverage area, depending on the context in which the term is used.

(31) A base station **110** may provide communication coverage for a macro cell, a pico cell, a femto cell, and/or another type of cell. A macro cell may cover a relatively large geographic area (e.g., several kilometers in radius) and may allow unrestricted access by UEs **120** with service subscriptions. A pico cell may cover a relatively small geographic area and may allow unrestricted access by UEs **120** with service subscription. A femto cell may cover a relatively small geographic area (e.g., a home) and may allow restricted access by UEs **120** having association with the femto cell (e.g., UEs **120** in a closed subscriber group (CSG)). A base station **110** for a macro cell may be referred to as a macro base station. A base station **110** for a pico cell may be referred to as a pico base station. A base station **110** for a femto cell may be referred to as a femto base station or an in-home base station. In the example shown in FIG. **1**, the BS **110***a* may be a macro base station for a macro cell **102***a*, the BS **110***b* may be a pico base station for a pico cell **102***b*, and the BS **110***c* may be a femto base station for a femto cell **102***c*. A base station may support one or multiple (e.g., three) cells.

(32) In some examples, a cell may not necessarily be stationary, and the geographic area of the cell may move according to the location of a base station **110** that is mobile (e.g., a mobile base station). In some examples, the base stations **110** may be interconnected to one another and/or to one or more other base stations **110** or network entities in the wireless network **100** through various types of backhaul interfaces, such as a direct physical connection or a virtual network, using any suitable transport network.
(33) In some aspects, the term “base station” (e.g., the base station **110**) or “network entity” may refer to an aggregated base station, a disaggregated base station, an integrated access and backhaul (IAB) node, a relay node, and/or one or more components thereof. For example, in some aspects, “base station” or “network entity” may refer to a central unit (CU), a distributed unit (DU), a radio unit (RU), a Near-Real Time (Near-RT) RAN Intelligent Controller (MC), or a Non-Real Time (Non-RT) RIC, or a combination thereof. In some aspects, the term “base station” or “network entity” may refer to one device configured to perform one or more functions, such as those described herein in connection with the base station **110**. In some aspects, the term “base station” or “network entity” may refer to a plurality of devices configured to perform the one or more functions. For example, in some distributed systems, each of a number of different devices (which may be located in the same geographic location or in different geographic locations) may be configured to perform at least a portion of a function, or to duplicate performance of at least a portion of the function, and the term “base station” or “network entity” may refer to any one or more of those different devices. In some aspects, the term “base station” or “network entity” may refer to one or more virtual base stations and/or one or more virtual base station functions. For example, in some aspects, two or more base station functions may be instantiated on a single device. In some aspects, the term “base station” or “network entity” may refer to one of the base station functions and not another. In this way, a single device may include more than one base station.
(34) The wireless network **100** may include one or more relay stations. A relay station is a network entity that can receive a transmission of data from an upstream station (e.g., a network entity or a UE **120**) and send a transmission of the data to a downstream station (e.g., a UE **120** or a network entity). A relay station may be a UE **120** that can relay transmissions for other UEs **120**. In the example shown in FIG. **1**, the BS **110***d* (e.g., a relay base station) may communicate with the BS **110***a* (e.g., a macro base station) and the UE **120***d* in order to facilitate communication between the BS **110***a* and the UE **120***d*. A base station **110** that relays communications may be referred to as a relay station, a relay base station, a relay, or the like.
(35) The wireless network **100** may be a heterogeneous network with network entities that include different types of BSs, such as macro base stations, pico base stations, femto base stations, relay base stations, or the like. These different types of base stations **110** may have different transmit power levels, different coverage areas, and/or different impacts on interference in the wireless network **100**. For example, macro base stations may have a high transmit power level (e.g., 5 to 40 watts) whereas pico base stations, femto base stations, and relay base stations may have lower transmit power levels (e.g., 0.1 to 2 watts).
(36) A network controller **130** may couple to or communicate with a set network entities and may provide coordination and control for these network entities. The network controller **130** may communicate with the base stations **110** via a backhaul communication link. The network entities may communicate with one another directly or indirectly via a wireless or wireline backhaul communication link.
(37) In some aspects, as shown, a cell may be provided by a network entity (e.g., base station **110**) of an NTN. As used herein, “non-terrestrial network” may refer to a network for which access is provided by a non-terrestrial base station, such as a base station carried by a satellite, a balloon, a dirigible, an airplane, an unmanned aerial vehicle, and/or a high altitude platform station. A network entity in an NTN (NTN entity) may use a polarization. For example, a network entity in a

satellite **135** (NTN entity) may transmit a communication to the UE **120** using a circular polarization **136** or a linear polarization **138**. Circular polarization occurs when the tip of the electric field of an electromagnetic wave at a fixed point in space traces a circle, and the electromagnetic wave may be formed by superposing two orthogonal linearly polarized waves of equal amplitude and a 90-degree phase difference. A circular polarization may be a right-hand circular polarization (RHCP) or a left-hand circular polarization (LHCP). Linear polarization occurs when the tip of the electric field of an electromagnetic wave at a fixed point in space oscillates along a straight line over time.

(38) The UEs **120** may be dispersed throughout the wireless network **100**, and each UE **120** may be stationary or mobile. A UE **120** may include, for example, an access terminal, a terminal, a mobile station, and/or a subscriber unit. A UE **120** may be a cellular phone (e.g., a smart phone), a personal digital assistant (PDA), a wireless modem, a wireless communication device, a handheld device, a laptop computer, a cordless phone, a wireless local loop (WLL) station, a tablet, a camera, a gaming device, a netbook, a smartbook, an ultrabook, a medical device, a biometric device, a wearable device (e.g., a smart watch, smart clothing, smart glasses, a smart wristband, smart jewelry (e.g., a smart ring or a smart bracelet)), an entertainment device (e.g., a music device, a video device, and/or a satellite radio), a vehicular component or sensor, a smart meter/sensor, industrial manufacturing equipment, a global positioning system device, and/or any other suitable device that is configured to communicate via a wireless medium.

(39) Some UEs **120** may be considered machine-type communication (MTC) or evolved or enhanced machine-type communication (eMTC) UEs. An MTC UE and/or an eMTC UE may include, for example, a robot, a drone, a remote device, a sensor, a meter, a monitor, and/or a location tag, that may communicate with a network entity, another device (e.g., a remote device), or some other entity. Some UEs **120** may be considered Internet-of-Things (IoT) devices, and/or may be implemented as NB-IoT (narrowband IoT) devices. Some UEs **120** may be considered a Customer Premises Equipment. A UE **120** may be included inside a housing that houses components of the UE **120**, such as processor components and/or memory components. In some examples, the processor components and the memory components may be coupled together. For example, the processor components (e.g., one or more processors) and the memory components (e.g., a memory) may be operatively coupled, communicatively coupled, electronically coupled, and/or electrically coupled.

(40) In general, any number of wireless networks **100** may be deployed in a given geographic area. Each wireless network **100** may support a particular RAT and may operate on one or more frequencies. A RAT may be referred to as a radio technology, an air interface, or the like. A frequency may be referred to as a carrier, a frequency channel, or the like. Each frequency may support a single RAT in a given geographic area in order to avoid interference between wireless networks of different RATs. In some cases, NR or 5G RAT networks may be deployed.

(41) In some examples, two or more UEs **120** (e.g., shown as UE **120***a* and UE **120***e*) may communicate directly using one or more sidelink channels (e.g., without using a network entity as an intermediary to communicate with one another). For example, the UEs **120** may communicate using peer-to-peer (P2P) communications, device-to-device (D2D) communications, a vehicle-to-everything (V2X) protocol (e.g., which may include a vehicle-to-vehicle (V2V) protocol, a vehicle-to-infrastructure (V2I) protocol, or a vehicle-to-pedestrian (V2P) protocol), and/or a mesh network. In such examples, a UE **120** may perform scheduling operations, resource selection operations, and/or other operations described elsewhere herein as being performed by the base station **110**.

(42) Devices of the wireless network **100** may communicate using the electromagnetic spectrum, which may be subdivided by frequency or wavelength into various classes, bands, channels, or the like. For example, devices of the wireless network **100** may communicate using one or more operating bands. In 5G NR, two initial operating bands have been identified as frequency range designations FR1 (410 MHz-7.125 GHz) and FR2 (24.25 GHz-52.6 GHz). It should be understood

that although a portion of FR1 is greater than 6 GHz, FR1 is often referred to (interchangeably) as a "Sub-6 GHz" band in various documents and articles. A similar nomenclature issue sometimes occurs with regard to FR2, which is often referred to (interchangeably) as a "millimeter wave" band in documents and articles, despite being different from the extremely high frequency (EHF) band (30 GHz-300 GHz) which is identified by the International Telecommunications Union (ITU) as a "millimeter wave" band.

(43) The frequencies between FR1 and FR2 are often referred to as mid-band frequencies. Recent 5G NR studies have identified an operating band for these mid-band frequencies as frequency range designation FR3 (7.125 GHz-24.25 GHz). Frequency bands falling within FR3 may inherit FR1 characteristics and/or FR2 characteristics, and thus may effectively extend features of FR1 and/or FR2 into mid-band frequencies. In addition, higher frequency bands are currently being explored to extend 5G NR operation beyond 52.6 GHz. For example, three higher operating bands have been identified as frequency range designations FR4a or FR4-1 (52.6 GHz-71 GHz), FR4 (52.6 GHz-114.25 GHz), and FR5 (114.25 GHz-300 GHz). Each of these higher frequency bands falls within the EHF band.

(44) With the above examples in mind, unless specifically stated otherwise, it should be understood that the term "sub-6 GHz" or the like, if used herein, may broadly represent frequencies that may be less than 6 GHz, may be within FR1, or may include mid-band frequencies. Further, unless specifically stated otherwise, it should be understood that the term "millimeter wave" or the like, if used herein, may broadly represent frequencies that may include mid-band frequencies, may be within FR2, FR4, FR4-a or FR4-1, and/or FR5, or may be within the EHF band. It is contemplated that the frequencies included in these operating bands (e.g., FR1, FR2, FR3, FR4, FR4-a, FR4-1, and/or FR5) may be modified, and techniques described herein are applicable to those modified frequency ranges.

(45) In some aspects, a wireless communication device (e.g., UE **120**, network entity) may include a communication manager **140** or **150**. As described in more detail elsewhere herein, the communication manager **140** or **150** may obtain, at an AS layer, an indication of an ISL activation that is to take place at an activation time. The communication manager **140** or **150** may receive communications and buffer the communications in a buffer at the AS layer. The communication manager **140** or **150** may release the communications from the buffer to an upper protocol layer at a rate of release that is to change between a transition start time and the activation time. Additionally, or alternatively, the communication manager **140** or **150** may perform one or more other operations described herein.

(46) In some aspects, the UE **120** may include a communication manager **140**. As described in more detail elsewhere herein, the communication manager **140** may obtain, at an AS layer, an indication of an ISL deactivation that is to take place at a deactivation time. The communication manager **140** may buffer communications in a buffer and release the communications from the buffer at a rate of release that is to change between the deactivation time and a transition end time; and transmit the communications. Additionally, or alternatively, the communication manager **140** may perform one or more other operations described herein.

(47) As indicated above, FIG. **1** is provided as an example. Other examples may differ from what is described with regard to FIG. **1**.

(48) FIG. **2** is a diagram illustrating an example **200** of a network entity (e.g., base station **110**) in communication with a UE **120** in a wireless network **100**, in accordance with the present disclosure. The base station **110** may be equipped with a set of antennas **234***a* through **234***t*, such as T antennas (T≥1). The UE **120** may be equipped with a set of antennas **252***a* through **252***r*, such as R antennas (R≥1).

(49) At the base station **110**, a transmit processor **220** may receive data, from a data source **212**, intended for the UE **120** (or a set of UEs **120**). The transmit processor **220** may select one or more modulation and coding schemes (MCSs) for the UE **120** based at least in part on one or more

channel quality indicators (CQIs) received from that UE **120**. The base station **110** may process (e.g., encode and modulate) the data for the UE **120** based at least in part on the MC S(s) selected for the UE **120** and may provide data symbols for the UE **120**. The transmit processor **220** may process system information (e.g., for semi-static resource partitioning information (SRPI)) and control information (e.g., CQI requests, grants, and/or upper layer signaling) and provide overhead symbols and control symbols. The transmit processor **220** may generate reference symbols for reference signals (e.g., a cell-specific reference signal (CRS) or a demodulation reference signal (DMRS)) and synchronization signals (e.g., a primary synchronization signal (PSS) or a secondary synchronization signal (SSS)). A transmit (TX) multiple-input multiple-output (MIMO) processor **230** may perform spatial processing (e.g., precoding) on the data symbols, the control symbols, the overhead symbols, and/or the reference symbols, if applicable, and may provide a set of output symbol streams (e.g., T output symbol streams) to a corresponding set of modems **232** (e.g., T modems), shown as modems **232***a* through **232***t*. For example, each output symbol stream may be provided to a modulator component (shown as MOD) of a modem **232**. Each modem **232** may use a respective modulator component to process a respective output symbol stream (e.g., for OFDM) to obtain an output sample stream. Each modem **232** may further use a respective modulator component to process (e.g., convert to analog, amplify, filter, and/or upconvert) the output sample stream to obtain a downlink signal. The modems **232***a* through **232***t* may transmit a set of downlink signals (e.g., T downlink signals) via a corresponding set of antennas **234** (e.g., T antennas), shown as antennas **234***a* through **234***t*. The base station **110** may be an NTN network entity located in a terrestrial location or in a non-terrestrial location (e.g., satellite **135**).

(50) At the UE **120**, a set of antennas **252** (shown as antennas **252***a* through **252***r*) may receive the downlink signals from the base station **110** and/or other base stations **110** and may provide a set of received signals (e.g., R received signals) to a set of modems **254** (e.g., R modems), shown as modems **254***a* through **254***r*. For example, each received signal may be provided to a demodulator component (shown as DEMOD) of a modem **254**. Each modem **254** may use a respective demodulator component to condition (e.g., filter, amplify, downconvert, and/or digitize) a received signal to obtain input samples. Each modem **254** may use a demodulator component to further process the input samples (e.g., for OFDM) to obtain received symbols. A MIMO detector **256** may obtain received symbols from the modems **254**, may perform MIMO detection on the received symbols if applicable, and may provide detected symbols. A receive processor **258** may process (e.g., demodulate and decode) the detected symbols, may provide decoded data for the UE **120** to a data sink **260**, and may provide decoded control information and system information to a controller/processor **280**. The term “controller/processor” may refer to one or more controllers, one or more processors, or a combination thereof. A channel processor may determine a reference signal received power (RSRP) parameter, a received signal strength indicator (RSSI) parameter, a reference signal received quality (RSRQ) parameter, and/or a CQI parameter, among other examples. In some examples, one or more components of the UE **120** may be included in a housing **284**.

(51) The network controller **130** may include a communication unit **294**, a controller/processor **290**, and a memory **292**. The network controller **130** may include, for example, one or more devices in a core network. The network controller **130** may communicate with the network entity via the communication unit **294**.

(52) One or more antennas (e.g., antennas **234***a* through **234***t* and/or antennas **252***a* through **252***r*) may include, or may be included within, one or more antenna panels, one or more antenna groups, one or more sets of antenna elements, and/or one or more antenna arrays, among other examples. An antenna panel, an antenna group, a set of antenna elements, and/or an antenna array may include one or more antenna elements (within a single housing or multiple housings), a set of coplanar antenna elements, a set of non-coplanar antenna elements, and/or one or more antenna elements coupled to one or more transmission and/or reception components, such as one or more

components of FIG. **2**.

(53) On the uplink, at the UE **120**, a transmit processor **264** may receive and process data from a data source **262** and control information (e.g., for reports that include RSRP, RSSI, RSRQ, and/or CQI) from the controller/processor **280**. The transmit processor **264** may generate reference symbols for one or more reference signals. The symbols from the transmit processor **264** may be precoded by a TX MIMO processor **266** if applicable, further processed by the modems **254** (e.g., for DFT-s-OFDM or CP-OFDM), and transmitted to the network entity. In some examples, the modem **254** of the UE **120** may include a modulator and a demodulator. In some examples, the UE **120** includes a transceiver. The transceiver may include any combination of the antenna(s) **252**, the modem(s) **254**, the MIMO detector **256**, the receive processor **258**, the transmit processor **264**, and/or the TX MIMO processor **266**. The transceiver may be used by a processor (e.g., the controller/processor **280**) and the memory **282** to perform aspects of any of the methods described herein (e.g., with reference to FIGS. **4-17**).

(54) At the network entity (e.g., base station **110**), the uplink signals from UE **120** and/or other UEs may be received by the antennas **234**, processed by the modem **232** (e.g., a demodulator component, shown as DEMOD, of the modem **232**), detected by a MIMO detector **236** if applicable, and further processed by a receive processor **238** to obtain decoded data and control information sent by the UE **120**. The receive processor **238** may provide the decoded data to a data sink **239** and provide the decoded control information to the controller/processor **240**. The network entity may include a communication unit **244** and may communicate with the network controller **130** via the communication unit **244**. The network entity may include a scheduler **246** to schedule one or more UEs **120** for downlink and/or uplink communications. In some examples, the modem **232** of the network entity may include a modulator and a demodulator. In some examples, the network entity includes a transceiver. The transceiver may include any combination of the antenna(s) **234**, the modem(s) **232**, the MIMO detector **236**, the receive processor **238**, the transmit processor **220**, and/or the TX MIMO processor **230**. The transceiver may be used by a processor (e.g., the controller/processor **240**) and the memory **242** to perform aspects of any of the methods described herein (e.g., with reference to FIGS. **4-17**).

(55) A controller/processor of a network entity, (e.g., the controller/processor **240** of the base station **110**), the controller/processor **280** of the UE **120**, and/or any other component(s) of FIG. **2** may perform one or more techniques associated with a gradual rate of release for a buffer in association with an ISL transition, as described in more detail elsewhere herein. In some aspects, the NTN entity is a network entity at the surface or at a satellite (e.g., satellite **135**). For example, the controller/processor **240** of the base station **110**, the controller/processor **280** of the UE **120**, and/or any other component(s) of FIG. **2** may perform or direct operations of, for example, process **1000** of FIG. **10**, process **1100** of FIG. **11**, and/or other processes as described herein. The memory **242** and the memory **282** may store data and program codes for the network entity and the UE **120**, respectively. In some examples, the memory **242** and/or the memory **282** may include a non-transitory computer-readable medium storing one or more instructions (e.g., code and/or program code) for wireless communication. For example, the one or more instructions, when executed (e.g., directly, or after compiling, converting, and/or interpreting) by one or more processors of the network entity and/or the UE **120**, may cause the one or more processors, the UE **120**, and/or the network entity to perform or direct operations of, for example, process **1000** of FIG. **10**, process **1100** of FIG. **11**, and/or other processes as described herein. In some examples, executing instructions may include running the instructions, converting the instructions, compiling the instructions, and/or interpreting the instructions, among other examples.

(56) In some aspects, a wireless communication device (e.g., UE **120**, network entity) includes means for obtaining, at an AS layer, an indication of an ISL activation that is to take place at an activation time; means for receiving communications; means for buffering the communications in a buffer at the AS layer; and/or means for releasing the communications from the buffer to an upper

protocol layer at a rate of release that changes between a transition start time and the activation time. In some aspects, the means for the wireless communication device to perform operations described herein may include, for example, one or more of communication manager **150**, transmit processor **220**, TX MIMO processor **230**, modem **232**, antenna **234**, MIMO detector **236**, receive processor **238**, controller/processor **240**, memory **242**, or scheduler **246**. In some aspects, the means for the wireless communication device to perform operations described herein may include, for example, one or more of communication manager **140**, antenna **252**, modem **254**, MIMO detector **256**, receive processor **258**, transmit processor **264**, TX MIMO processor **266**, controller/processor **280**, or memory **282**.

(57) In some aspects, the UE **120** includes means for obtaining, at an AS layer, an indication of an ISL deactivation that is to take place at a deactivation time; means for buffering communications in a buffer; means for releasing the communications from the buffer at a rate of release that changes between the deactivation time and a transition end time; and/or means for transmitting the communications. The means for the UE **120** to perform operations described herein may include, for example, one or more of communication manager **140**, antenna **252**, modem **254**, MIMO detector **256**, receive processor **258**, transmit processor **264**, TX MIMO processor **266**, controller/processor **280**, or memory **282**.

(58) As indicated above, FIG. **2** is provided as an example. Other examples may differ from what is described with regard to FIG. **2**.

(59) FIG. **3** is a diagram illustrating an example of a disaggregated base station **300**, in accordance with the present disclosure.

(60) Deployment of communication systems, such as 5G NR systems, may be arranged in multiple manners with various components or constituent parts. In a 5G NR system, or network, a network node, a network entity, a mobility element of a network, a radio access network (RAN) node, a core network node, a network element, or a network equipment, such as a base station, or one or more units (or one or more components) performing base station functionality, may be implemented in an aggregated or disaggregated architecture. For example, a BS (such as a Node B, evolved NB (eNB), NR BS, 5G NB, access point (AP), a TRP, or a cell, etc.) may be implemented as an aggregated base station (also known as a standalone BS or a monolithic BS) or a disaggregated base station.

(61) An aggregated base station may be configured to utilize a radio protocol stack that is physically or logically integrated within a single RAN node. A disaggregated base station may be configured to utilize a protocol stack that is physically or logically distributed among two or more units (such as one or more CUs, one or more DUs, or one or more RUs). In some aspects, a CU may be implemented within a RAN node, and one or more DUs may be co-located with the CU, or alternatively, may be geographically or virtually distributed throughout one or multiple other RAN nodes. The DUs may be implemented to communicate with one or more RUs. Each of the CU, DU and RU also can be implemented as virtual units (e.g., a virtual central unit (VCU), a virtual distributed unit (VDU), or a virtual radio unit (VRU)).

(62) Base station-type operation or network design may consider aggregation characteristics of base station functionality. For example, disaggregated base stations may be utilized in an IAB network, an open radio access network (O-RAN (such as the network configuration sponsored by the O-RAN Alliance)), or a virtualized radio access network (vRAN, also known as a cloud radio access network (C-RAN)). Disaggregation may include distributing functionality across two or more units at various physical locations, as well as distributing functionality for at least one unit virtually, which can enable flexibility in network design. The various units of the disaggregated base station, or disaggregated RAN architecture, can be configured for wired or wireless communication with at least one other unit.

(63) The disaggregated base station **300** architecture may include one or more CUs **310** that can communicate directly with a core network **320** via a backhaul link, or indirectly with the core

network **320** through one or more disaggregated base station units (such as a Near-RT RIC **325** via an E2 link, or a Non-Real Time (Non-RT) MC **315** associated with a Service Management and Orchestration (SMO) Framework **305**, or both). A CU **310** may communicate with one or more DUs **330** via respective midhaul links, such as an F1 interface. The DUs **330** may communicate with one or more RUs **340** via respective fronthaul links. The fronthaul link, the midhaul link, and the backhaul link may be generally referred to as “communication links.” The RUs **340** may communicate with respective UEs **120** via one or more RF access links. In some aspects, the UE **120** may be simultaneously served by multiple RUs **340**. The DUs **330** and the RUs **340** may also be referred to as “O-RAN DUs (O-DUs”) and “O-RAN RUs (O-RUs)”, respectively. A network entity may include a CU, a DU, an RU, or any combination of CUs, DUs, and RUs. A network entity may include a disaggregated base station or one or more components of the disaggregated base station, such as a CU, a DU, an RU, or any combination of CUs, DUs, and RUs. A network entity may also include one or more of a TRP, a relay station, a passive device, an intelligent reflective surface (IRS), or other components that may provide a network interface for or serve a UE, mobile station, sensor/actuator, or other wireless device.

(64) Each of the units, i.e., the CUs **310**, the DUs **330**, the RUs **340**, as well as the Near-RT RICs **325**, the Non-RT RICs **315** and the SMO Framework **305**, may include one or more interfaces or be coupled to one or more interfaces configured to receive or transmit signals, data, or information (collectively, signals) via a wired or wireless transmission medium. Each of the units, or an associated processor or controller providing instructions to the communication interfaces of the units, can be configured to communicate with one or more of the other units via the transmission medium. For example, the units can include a wired interface configured to receive or transmit signals over a wired transmission medium to one or more of the other units. Additionally, the units can include a wireless interface, which may include a receiver, a transmitter or transceiver (such as an RF transceiver), configured to receive or transmit signals, or both, over a wireless transmission medium to one or more of the other units.

(65) In some aspects, the CU **310** may host one or more higher layer control functions. Such control functions can include radio resource control (RRC), packet data convergence protocol (PDCP), service data adaptation protocol (SDAP), or the like. Each control function can be implemented with an interface configured to communicate signals with other control functions hosted by the CU **310**. The CU **310** may be configured to handle user plane functionality (i.e., Central Unit-User Plane (CU-UP)), control plane functionality (i.e., Central Unit-Control Plane (CU-CP)), or a combination thereof. In some implementations, the CU **310** can be logically split into one or more CU-UP units and one or more CU-CP units. The CU-UP unit can communicate bidirectionally with the CU-CP unit via an interface, such as the E1 interface when implemented in an O-RAN configuration. The CU **310** can be implemented to communicate with the DU **330**, as necessary, for network control and signaling.

(66) The DU **330** may correspond to a logical unit that includes one or more base station functions to control the operation of one or more RUs **340**. In some aspects, the DU **330** may host one or more of a radio link control (RLC) layer, a medium access control (MAC) layer, and one or more high physical (PHY) layers (such as modules for forward error correction (FEC) encoding and decoding, scrambling, modulation and demodulation, or the like) depending, at least in part, on a functional split, such as those defined by the 3GPP. In some aspects, the DU **330** may further host one or more low PHY layers. Each layer (or module) can be implemented with an interface configured to communicate signals with other layers (and modules) hosted by the DU **330**, or with the control functions hosted by the CU **310**.

(67) Lower-layer functionality can be implemented by one or more RUs **340**. In some deployments, an RU **340**, controlled by a DU **330**, may correspond to a logical node that hosts RF processing functions, or low-PHY layer functions (such as performing fast Fourier transform (FFT), inverse FFT (iFFT), digital beamforming, physical random access channel (PRACH) extraction and

filtering, or the like), or both, based at least in part on the functional split, such as a lower layer functional split. In such an architecture, the RU(s) **340** can be implemented to handle over the air (OTA) communication with one or more UEs **120**. In some implementations, real-time and non-real-time aspects of control and user plane communication with the RU(s) **340** can be controlled by the corresponding DU **330**. In some scenarios, this configuration can enable the DU(s) **330** and the CU **310** to be implemented in a cloud-based RAN architecture, such as a vRAN architecture.
(68) The SMO Framework **305** may be configured to support RAN deployment and provisioning of non-virtualized and virtualized network elements. For non-virtualized network elements, the SMO Framework **305** may be configured to support the deployment of dedicated physical resources for RAN coverage requirements which may be managed via an operations and maintenance interface (such as an O1 interface). For virtualized network elements, the SMO Framework **305** may be configured to interact with a cloud computing platform (such as an open cloud (O-Cloud) **390**) to perform network element life cycle management (such as to instantiate virtualized network elements) via a cloud computing platform interface (such as an O2 interface). Such virtualized network elements can include, but are not limited to, CUs **310**, DUs **330**, RUs **340** and Near-RT RICs **325**. In some implementations, the SMO Framework **305** can communicate with a hardware aspect of a 4G RAN, such as an open eNB (O-eNB) **311**, via an O1 interface. Additionally, in some implementations, the SMO Framework **305** can communicate directly with one or more RUs **340** via an O1 interface. The SMO Framework **305** also may include a Non-RT RIC **315** configured to support functionality of the SMO Framework **305**.
(69) The Non-RT RIC **315** may be configured to include a logical function that enables non-real-time control and optimization of RAN elements and resources, Artificial Intelligence/Machine Learning (AI/ML) workflows including model training and updates, or policy-based guidance of applications/features in the Near-RT RIC **325**. The Non-RT RIC **315** may be coupled to or communicate with (such as via an A1 interface) the Near-RT RIC **325**. The Near-RT RIC **325** may be configured to include a logical function that enables near-real-time control and optimization of RAN elements and resources via data collection and actions over an interface (such as via an E2 interface) connecting one or more CUs **310**, one or more DUs **330**, or both, as well as an O-eNB, with the Near-RT RIC **325**.
(70) In some implementations, to generate AI/ML models to be deployed in the Near-RT RIC **325**, the Non-RT RIC **315** may receive parameters or external enrichment information from external servers. Such information may be utilized by the Near-RT RIC **325** and may be received at the SMO Framework **305** or the Non-RT RIC **315** from non-network data sources or from network functions. In some examples, the Non-RT RIC **315** or the Near-RT RIC **325** may be configured to tune RAN behavior or performance. For example, the Non-RT RIC **315** may monitor long-term trends and patterns for performance and employ AI/ML models to perform corrective actions through the SMO Framework **305** (such as reconfiguration via O1) or via creation of RAN management policies (such as A1 policies).
(71) As indicated above, FIG. **3** is provided as an example. Other examples may differ from what is described with regard to FIG. **3**.
(72) FIGS. **4**A-**4**C are diagrams illustrating examples of a satellite deployments in an NTN, in accordance with the present disclosure.
(73) Example **400** in FIG. **4**A shows a regenerative satellite deployment. In example **400**, a UE **120** is served by a satellite **420** (e.g., satellite **135**) via a service link **430**. For example, the satellite **420** may include a network entity (e.g., a base station **110**, BS **110***a*, a gNB). In some examples, the satellite **420** may be referred to as a non-terrestrial base station, a regenerative repeater, an on-board processing repeater, and/or an NTN entity. In some examples, the satellite **420** may demodulate an uplink radio frequency signal and may modulate a baseband signal derived from the uplink radio signal to produce a downlink radio frequency transmission. The satellite **420** may transmit the downlink radio frequency signal on the service link **430**. The satellite **420** may provide

a cell that covers the UE **120**. In some examples, the satellite **420** may further communicate with a ground gateway via a feeder link. The satellite **420** may connect to a RAN, a core network, and/or a data network through the gateway.

(74) Example **410** in FIG. **4**B shows a transparent satellite deployment, which may also be referred to as a bent-pipe satellite deployment. In example **410**, a UE **120** is served by a satellite **440** via the service link **430**. The satellite **440** may also be considered to be an NTN entity. The NTN entity may include a CU, one or more DUs, and/or one or more RUs, depending on the deployment with other NTN entities and ground stations. The satellite **440** may be a transparent satellite. The satellite **440** may relay a signal received from the UE **120** via a service link **430** to the gateway **450** via a feeder link **460**. In another example, the satellite **440** may receive an uplink radio frequency transmission from the service link **430** and may transmit an uplink radio frequency transmission to the feeder link **460** without demodulating the uplink radio frequency transmission. In some examples, the satellite may frequency convert the uplink radio frequency transmission received on the service link **430** to a frequency of the uplink radio frequency transmission on the feeder link **460** and may amplify and/or filter the uplink radio frequency transmission. In some examples, similar as in the uplink direction, the satellite **440** may relay a downlink signal received from the gateway **450** via the feeder link **460** to the UE **120** via the service link **430**. In some examples, the UEs **120** shown in example **400** and example **410** may be associated with a Global Navigation Satellite System (GNSS) capability, a Global Positioning System (GPS) capability, and/or the like, though not all UEs have such capabilities. The satellite **440** may provide a cell that covers the UE **120**.

(75) The service link **430** may include a link between the satellite **440** and the UE **120** and may include one or more of an uplink or a downlink. The feeder link **460** may include a link between the satellite **440** and the gateway **450** and may include one or more of an uplink (e.g., from the UE **120** to the gateway **450**) or a downlink (e.g., from the gateway **450** to the UE **120**).

(76) The feeder link **460** and the service link **430** may each experience Doppler effects due to the movement of the satellites **420** and **440** and, potentially, movement of a UE **120**. These Doppler effects may be significantly larger than in a terrestrial network. The Doppler effect on the feeder link **460** may be compensated to some degree but may still be associated with some amount of uncompensated frequency error. Furthermore, the gateway **450** may be associated with a residual frequency error, and/or the satellite **420**/**440** may be associated with an on-board frequency error. These sources of frequency error may cause a received downlink frequency at the UE **120** to drift from a target downlink frequency.

(77) Satellites **420** and **440** may be a satellite in a GSO or GEO, which may be, for example, 36,000 kilometers (km) above the Earth. The speed of the satellite with respect to Earth may be negligible but have a round-trip propagation delay of more than 500 milliseconds (ms), as compared to 25 ms for an LEO satellite at 600 km above the Earth. In an NTN, where a distance between UE **120** and a satellite can be larger than 600 km, a pathloss change may not be appropriately reflected in a propagation delay change. The UE **120** is expected to be able to autonomously pre-compensate for propagation delay all the way to a reference point, and thus timing advance (TA) validation can be carried out more directly rather than relying on indirect parameters such as RSRP. The UE **120** may use a TA for timing alignment of communications due to a propagation delay. The TA may inform the UE **120** to transmit a communication in uplink earlier than a reference downlink slot timing by the TA amount. The TA may need to be validated if the propagation distance between the UE **120** and the network entity changes. In addition, a satellite may not be always available to the UE **120** if the satellite is in an NGSO.

(78) Some satellite operators may deploy satellite access with intentional coverage gaps, or discontinuous coverage (DC). DC may involve LEO satellites, IoT networks, UAVs, a HAPS, or a satellite constellation that does not cover a particular location on Earth all of the time. Some services may be delay tolerant and may use DC. Such services may include services for IoT

devices and networks, services for utility meters, or services that use sensors. DC may be used to address service availability and power consumption. In addition to DC, the feeder link **460** may have intermittent connectivity with a ground station, such as gateway **450**. There may be some areas where it is not feasible to deploy a ground station, either due to deployment designs and/or costs (e.g., in remote geographical locations with small populations).
(79) To deal with unavailability of the feeder link **460**, a satellite transmitter may include network elements at the satellite **420**/**440**. Both the radio network and the core network aspects may be mounted on the satellite **420**/**440**. In one scenario, the entire control plane and user plane paths may be mounted on the satellite **420**/**440**. For example, the control plane management entity (e.g., mobile management entity (MME) in 4G or an access and mobility management function (AMF) and a session management function (SMF) in NR) as well as the user plane path (e.g., serving gateway (GW) and packet data network GW for 4G or a user plane function in NR) may be mounted on the satellite **420**/**440**. In this scenario, the ground station (e.g., gateway **450**) may include the home subscriber services (HSS) including subscription management, user authentication and authorization. Also, in this scenario, some portions of the control plane path can be at the ground station (e.g., coordination between the control plane management entities in different satellites, buffering of user data and control plane data while the feeder link is unavailable). In this scenario, there may also be user data plane and control plane data buffering at the satellite due to DC. Finally, the internet connection for the data plane may be at the ground station. Other functional splits between the satellite transmitter, and the ground station may be employed.
(80) Example **470** in FIG. **4**C shows use of an ISL **480** between satellite **440** and satellite **490**. The ISL **480** may be used to support coverage extension in the NTN. For example, if gateway **450** is deployed with certain limitations and satellite **440** does not have a direct feeder link connection to gateway **450**, satellite **440** may be connected with the ISL **480** to satellite **490**, which has the feeder link **460**. In this way, satellite **440** has access to the gateway **450** and service availability improves.
(81) In some aspects, network functions of a disaggregated base station may be split between satellites. For example, satellite **440** may perform functions of a DU and satellite **490** may perform functions of a CU.
(82) As indicated above, FIGS. **4**A-**4**C provide some examples. Other examples may differ from what is described with regard to FIGS. **4**A-**4**C.
(83) FIGS. **5**A and **5**B are diagrams illustrating an example **500** of ISL activation and an example **502** of ISL deactivation, respectively, in accordance with the present disclosure.
(84) Example **500** in FIG. **5**A shows that satellite **440** may have a feeder link **510**. However, a topology change event may occur, where there is a change in a communication path in the NTN topology. This may include a change in a link that is used or a change in an operation of an NTN entity. In example **500**, the topology change event is an ISL activation, where satellite **440** no longer uses feeder link **510** and switches to use of the ISL **480** to peer satellite **490**. This may be due to satellite **440** moving to a location where there is no line of sight to the gateway **450**. From the viewpoint of the UE, the connection or communication path between satellite **440** and the gateway **450** may be an extended feeder link, even though the communication path proceeds through the ISL **480** and satellite **490**.
(85) Example **502** in FIG. **5**B shows that a topology change event may be an ISL deactivation, where satellite switches from using the ISL **480** to using the feeder link **510**. For example, satellite **440** or the network may determine that satellite **440** has or will have a line of sight with the gateway **450**. In other words, satellite **440** may activate use of the ISL **480** when moving further away from the gateway **450** and deactivate use of the ISL **480** when moving closer to the gateway **450**.
(86) As indicated above, FIGS. **5**A and **5**B are provided as examples. Other examples may differ from what is described with regard to FIGS. **5**A and **5**B.

(87) FIG. **6** is a diagram illustrating an example **600** of a change in propagation delays associated with an ISL, in accordance with the present disclosure.
(88) When the ISL **480** is activated or deactivated, the signal propagation delay between the UE **120** and the gateway **450** changes because the first propagation delay **602** (T1+T2) for signal propagation through satellite **440** is less than the second propagation delay **604** (T1+T3+T4) for signal propagation through both satellite **440** and satellite **490** via ISL **480**. Accordingly, the communication service will experience a sudden change in latency. The latency change **606** at the ISL activation (time t0) is the difference between the first propagation delay **602** and the second propagation delay **604**. According to one or more examples, if satellite **440** is an LEO satellite and satellite **490** is a GEO satellite, the latency change **606** may be over 200 ms. The sudden change in latency can impact the UE's communication service because the UE's communication service is not aware of the reason for such a sudden change. The sudden change in latency may cause unexpected communication service behavior, such as a communication service timeout (e.g., transmission control protocol (TCP) spurious timeout when the round-trip-time (RTT) exceeds the retransmission timer), triggering of a congestion control algorithm (e.g., web real-time communication (WebRTC) algorithm that controls congestion in compliance with the WebRTC framework used for browsers and video calls), or change in the quality of the communication service experience (e.g., quality of experience (QoE) change).
(89) According to one or more aspects described herein, instead of an immediate change in the propagation delay, a UE (e.g., UE **120**) may gradually increase or decrease the latency between a first propagation delay before a topology change event (e.g., satellite switch, activation of ISL, deactivation of ISL) and a second propagation delay after the topology change event. This may include buffering communications in a buffer in the memory (e.g., memory **282**) and changing a rate of release of communications from the buffer of the UE such that the change in the propagation delay experienced by a service application of the UE transitions gradually from the first propagation delay to the second propagation delay, rather than changing instantly. For the UE receiving communications in association with an ISL activation, the UE may release the communications from the buffer at a rate of release that changes between a transition start time and the ISL activation time. The transition start time may be an estimated time to start changing the rate of release. For the UE transmitting communications in association with an ISL deactivation, the UE may release the communications from the buffer at a rate of release that changes between the ISL deactivation time and a transition end time. The transition end time may be an estimated time to stop changing the rate of release. In this way, the UE may avoid communication timeouts, the triggering of traffic congestion algorithms, unexpected changes in the communication experience, or other results that may drop service or consume additional resources when reconnecting the service. As a result, the UE may reduce latency and conserve processing resources and signaling resources.
(90) As indicated above, FIG. **6** is provided as an example. Other examples may differ from what is described with regard to FIG. **6**.
(91) FIG. **7** is a diagram illustrating an example **700** of a user plane protocol stack and a control plane protocol stack for a network entity (e.g., base station **110**) and a core network in communication with a UE **120**, in accordance with the present disclosure. In some aspects, the network entity may include a plurality of network nodes. In some aspects, protocol stack functions of the network entity may be distributed across multiple network nodes. For example, a first network entity may implement a first layer of a protocol stack, and a second network entity may implement a second layer of the protocol stack. The distribution of the protocol stack across network nodes (in examples where the protocol stack is distributed across network nodes) may be based at least in part on a functional split, as described elsewhere herein. It should be understood that references to “a network node” or “the network node” can, in some aspects, refer to multiple network nodes.

(92) On the user plane, the UE **120** and the network entity may include respective PHY layers, MAC layers, RLC layers, PDCP layers, and SDAP layers. A user plane function may handle transport of user data between the UE **120** and the network entity. On the control plane, the UE **120** and the network entity may include respective RRC layers. Furthermore, the UE **120** may include a non-access stratum (NAS) layer in communication with an NAS layer of an AMF. The NAS layer may be a functional layer used to manage the establishment of communication sessions and to maintain continuous communications (in contrast to AS layer that is responsible for carrying information over the wireless portions of the network). The NAS uses a set of protocols to convey non-radio signaling between the UE and the 5G core. The AMF may be associated with a core network associated with the network entity, such as a 5G core network (5GC) or a next-generation radio access network (NG-RAN). A control plane function may handle transport of control information between the UE and the core network and/or between the UE and the NG-RAN. Generally, a first layer is referred to as higher than a second layer if the first layer is further from the PHY layer than the second layer. For example, the PHY layer may be referred to as a lowest layer, and the SDAP/PDCP/RLC/MAC layer may be referred to as higher than the PHY layer and lower than the RRC layer. An AS layer **702** may be any of the SDAP/PDCP/RLC/MAC layers. The AS layer **702** may be a layer that is involved with radio operations. An application (APP) layer may be an upper protocol layer **704** that is higher than the SDAP/PDCP/RLC/MAC layer. The upper protocol layer **704** may be a layer that manages applications or services. In some examples, the APP layer may reside in an entity belonging to the data network, instead of residing in the network entity (e.g., base station **110**). In some examples, an entity may handle the services and functions of a given layer (e.g., a PDCP entity may handle the services and functions of the PDCP layer), though the description herein refers to the layers themselves as handling the services and functions.
(93) The RRC layer may handle communications related to configuring and operating the UE **120**, such as: broadcast of system information related to the AS and the NAS; paging initiated by the 5GC or the NG-RAN; establishment, maintenance, and release of an RRC connection between the UE and the NG-RAN, including addition, modification, and release of carrier aggregation, as well as addition, modification, and release of dual connectivity; security functions including key management; establishment, configuration, maintenance, and release of signaling radio bearers (SRBs) and data radio bearers (DRBs); mobility functions (e.g., handover and context transfer, UE cell selection and reselection and control of cell selection and reselection, inter-RAT mobility); quality of service (QoS) management functions; UE measurement reporting and control of the reporting; detection of and recovery from radio link failure; and NAS message transfer between the NAS layer and the lower layers of the UE **120**. The RRC layer is frequently referred to as Layer 3 (L3).
(94) The SDAP layer, PDCP layer, RLC layer, and MAC layer may be collectively referred to as Layer 2 (L2). Thus, in some cases, the SDAP, PDCP, RLC, and MAC layers are referred to as sublayers of Layer 2. On the transmitting side (e.g., if the UE **120** is transmitting an uplink communication or the network node **110** is transmitting a downlink communication), the SDAP layer may receive a data flow in the form of a QoS flow. A QoS flow is associated with a QoS identifier, which identifies a QoS parameter associated with the QoS flow, and a QoS flow identifier (QFI), which identifies the QoS flow. Policy and charging parameters are enforced at the QoS flow granularity. A QoS flow can include one or more service data flows (SDFs), so long as each SDF of a QoS flow is associated with the same policy and charging parameters. In some aspects, the RRC/NAS layer may generate control information to be transmitted and may map the control information to one or more radio bearers for provision to the PDCP layer.
(95) The SDAP layer, or the RRC/NAS layer, may map QoS flows or control information to radio bearers. Thus, the SDAP layer may be said to handle QoS flows on the transmitting side. The SDAP layer may provide the QoS flows to the PDCP layer via the corresponding radio bearers. The PDCP layer may map radio bearers to RLC channels. The PDCP layer may handle various services

and functions on the user plane, including sequence numbering, header compression and decompression (if robust header compression is enabled), transfer of user data, reordering and duplicate detection (if in-order delivery to layers above the PDCP layer is required), PDCP protocol data unit (PDU) routing (in case of split bearers), retransmission of PDCP service data units (SDUs), ciphering and deciphering, PDCP SDU discard (e.g., in accordance with a timer, as described elsewhere herein), PDCP re-establishment and data recovery for RLC acknowledged mode (AM), and duplication of PDCP PDUs. The PDCP layer may handle similar services and functions on the control plane, including sequence numbering, ciphering, deciphering, integrity protection, transfer of control plane data, duplicate detection, and duplication of PDCP PDUs.
(96) The PDCP layer may provide data, in the form of PDCP PDUs, to the RLC layer via RLC channels. The RLC layer may handle transfer of upper layer PDUs to the MAC and/or PHY layers, sequence numbering independent of PDCP sequence numbering, error correction via automatic repeat requests (ARQ), segmentation and re-segmentation, reassembly of an SDU, RLC SDU discard, and RLC re-establishment.
(97) The RLC layer may provide data, mapped to logical channels, to the MAC layer. The services and functions of the MAC layer include mapping between logical channels and transport channels (used by the PHY layer as described below), multiplexing/demultiplexing of MAC SDUs belonging to one or different logical channels into/from transport blocks (TBs) delivered to/from the physical layer on transport channels, scheduling information reporting, error correction through hybrid ARQ (HARD), priority handling between UEs by means of dynamic scheduling, priority handling between logical channels of one UE by means of logical channel prioritization, and padding.
(98) The MAC layer may package data from logical channels into TBs, and may provide the TBs on one or more transport channels to the PHY layer. The PHY layer may handle various operations relating to transmission of a data signal, as described in more detail in connection with FIG. **2**. The PHY layer is frequently referred to as Layer 1 (L1).
(99) On the receiving side (e.g., if the UE **120** is receiving a downlink communication or the network entity is receiving an uplink communication), the operations may be similar to those described for the transmitting side, but reversed. For example, the PHY layer may receive TBs and may provide the TBs on one or more transport channels to the MAC layer. The MAC layer may map the transport channels to logical channels and may provide data to the RLC layer via the logical channels. The RLC layer may map the logical channels to RLC channels and may provide data to the PDCP layer via the RLC channels. The PDCP layer may map the RLC channels to radio bearers and may provide data to the SDAP layer or the RRC/NAS layer via the radio bearers.
(100) Data may be passed between the layers in the form of PDUs and SDUs. An SDU is a unit of data that has been passed from a layer or sublayer to a lower layer. For example, the PDCP layer may receive a PDCP SDU. A given layer may then encapsulate the unit of data into a PDU and may pass the PDU to a lower layer. For example, the PDCP layer may encapsulate the PDCP SDU into a PDCP PDU and may pass the PDCP PDU to the RLC layer. The RLC layer may receive the PDCP PDU as an RLC SDU, may encapsulate the RLC SDU into an RLC PDU, and so on. In effect, the PDU carries the SDU as a payload.
(101) In some aspects, a UE may operate one or more buffers at the AS layer **702**. The one or more buffers may be included in the memory **282**. For example, a buffer **706** in the memory **282** of a UE (e.g., UE **120**) may buffer (e.g., store in memory for later release from the buffer) received communications and later release the communications. The buffer **706** may exist or operate at the AS layer **702** (e.g., at an RLC layer) and the buffer **706** may release communications to the upper protocol layer **704**. An application that provides a communication service may exit or operate at the upper protocol layer **704** (e.g., application layer, a service layer, transport layer). The application may determine if there is a problem (e.g., loss of signal, too much delay, congestion) with a feeder link or a service link and take action to remedy the problem (e.g., reestablish a link). The buffer

706 may release communication to the upper protocol layer **704** to be used by the application.
(102) In some aspects, the buffer **706** may store communications to be transmitted. The buffer may exist or operate at the AS layer **702**. The buffer **706** may release the communications to a lower protocol layer, such as the MAC layer or the PHY layer for transmission.
(103) In some aspects, the buffer **706** may store communications at a rate (rate of buffering) that is used to assist with the gradual increase or decrease in latency. The rate of buffering may be based at least in part on a difference in propagation delays, a configured function, and/or a latency sensitivity of a service or quality of service flow.
(104) As indicated above, FIG. **7** is provided as an example. Other examples may differ from what is described with regard to FIG. **7**.
(105) FIG. **8** is a diagram illustrating an example **800** of a gradual increase in latency for received communications, in accordance with the present disclosure.
(106) Example **800** shows a gradual transition of the UE **820** from a transition start time **802** (t0) to an ISL activation time **804** (t1). The gradual transition may be part of a transition phase that occurs before ISL activation. The gradual transition involves a latency change **806** from a first propagation delay **808** (communication via NTN entity **810**) before ISL activation to a second propagation delay **812** (communication via NTN entity **810** and NTN entity **840**) after ISL activation.
(107) Example **800** also shows an NTN entity **810** (e.g., base station **110**, network entity, satellite **420**, satellite **440**) and a UE **820** (e.g., UE **120**) may communicate with one another. The UE **820** and a terrestrial network entity (e.g., ground station **830**) may communicate via the NTN entity **810**. The ground station **830** may be a base station (e.g., base station **110**) or a gateway (e.g., gateway **450**). There may be another NTN entity **840** (e.g., base station **110**, network entity, satellite **420**, satellite **490**) that has a feeder link (e.g., feeder link **460**) to the terrestrial network entity, and there may be an ISL (e.g., ISL **480**) that is activated or deactivated between NTN entity **810** and NTN entity **840**.
(108) As shown by reference number **845**, the UE **820** and the NTN entity **810** may communicate via the service link **430**. As shown by reference number **850**, the NTN entity **810** and the ground station **830** may communicate via feeder link **510**. This is before a topology change event such as the ISL activation.
(109) As shown by reference number **855**, the NTN entity **810** may transmit an indication of an ISL activation. The indication may be received in an SIB, which may be NTN-specific or dedicated for NTN. The indication may be received in downlink control information (DCI), a MAC CE, or an RRC message. The indication may be received via a short message over a physical downlink control channel (PDCCH) in a paging occasion (e.g., for RRC inactive or RRC idle UE). The indication may be received in a dedicated RRC message for RRC connected UE. For example, the indication may be a handover command, where the target cell identifier (ID) contained in the handover command indicates the same cell ID as the UE **820**'s current serving cell over the service link **430** before the topology change event. Thus, if the UE **820** receives a handover command that requests the UE **820** to hand over to the current serving cell, the UE **820** may detect the indication. The indication may be broadcast in a system information block (SIB). The indication may activate the buffer used for gradually increasing or decreasing latency.
(110) In some aspects, the UE **820** may obtain the indication through a schedule, a calculation, or another communication. The indication may include the ISL activation time **804**. The UE **820** may also receive an indication of the transition start time **802**. Alternatively, the UE **820** may calculate the transition start time **802** based at least in part on an indication or a calculated estimate of the latency change **806**. The UE **820** may calculate the transition start time **802** based at least in part on an indication of changes to a service link or a feeder link.
(111) As shown by reference number **860**, the UE **820** may buffer or may have buffered communications received from the NTN entity **810**. The UE **820** may buffer the communications in a buffer (e.g., buffer **706**) at the AS layer **702**. The buffer may be an additional buffer between the

AS layer **702** and the upper protocol layer **704**. The additional buffer may be a dedicated buffer that is activated only for the gradual increase or decrease of latency.
(112) As shown by reference number **865**, the UE **820** may release the communications at a rate of release **866** that changes between the transition start time **802** and the ISL activation time **804**. For example, the rate of release may be decreasing and thus the latency may be increasing. The buffer may release the communications to an upper protocol layer (e.g., upper protocol layer **704**, an application layer, a service layer, a transport layer). In some aspects, the rate of release **866** of the communications from the buffer may be based on a function of time, where a time instance or other time information (e.g., ISL activation time **804**, transition start time **802**, first propagation delay **808**, second propagation delay **812**, latency change **806**) are inputs, and an amount of latency and/or the rate of release **866** is the output. The function may be linear function that gradually increases at a steady rate. The function may be a stepped function where the latency increases a specified amount at instances of time. The function may be a non-linear function or any other function for changing the rate of release **866** between the transition start time **802** and the ISL activation time **804**. The UE **820** may change the rate of release by changing a timer **708** of the buffer or otherwise adjusting a buffering time. The UE **820** may change the rate of release or rate of buffering by adjusting the timer **708** for the buffer at a specific layer, such as at the RLC layer or the PDCP layer (e.g., t-Reassembly timer, t-Reordering timer).
(113) As shown by reference number **870**, NTN entity **810** may perform the ISL activation (shown by ISL activation **872**) at the ISL activation time **804**. By this time, the gradual increase in latency may have been completed. As a result, there is no sudden change in latency and communications are not disrupted. This reduces latency and conserves processing resources and signaling resources.
(114) While example **800** shows the latency increasing based on the rate of release **866** because the first propagation delay **808** before ISL activation is less than the second propagation delay **812** after ISL activation, in some aspects, the latency may be decreasing if the first propagation delay **808** before ISL activation is greater than the second propagation delay **812** after ISL activation.
(115) As shown by reference number **880**, the UE **820** and NTN entity **810** may communicate using the service link **430** shown in FIG. **5**. As shown by reference number **885**, NTN entity **810** may communicate with NTN entity **840** via the ISL **480**. As shown by reference number **890**, NTN entity **840** may communicate with the ground station **830** using the feeder link **510** shown in FIG. **5**.
(116) As indicated above, FIG. **8** is provided as an example. Other examples may differ from what is described with regard to FIG. **8**.
(117) FIG. **9** is a diagram illustrating an example **900** of a gradual increase in latency for received communications, in accordance with the present disclosure.
(118) Example **900** shows a gradual transition of the UE **820** from an ISL deactivation time **902** (t0) to a transition end time **904** (t1). The gradual transition may be part of a transition phase that occurs after ISL deactivation. The gradual transition involves a latency change **906** from a first propagation delay **908** (communication via NTN entity **810** and NTN entity **840**) before ISL deactivation to a second propagation delay **910** (communication via NTN entity **810**) after ISL deactivation. The latency may be decreasing at a rate of release **912** that is increasing.
(119) As shown by reference number **915**, the UE **820** and NTN entity **810** may communicate using the service link **430** shown in FIG. **5**. As shown by reference number **920**, NTN entity **810** may communicate with NTN entity **840** via the ISL **480**. As shown by reference number **925**, NTN entity **840** may communicate with the ground station **830** using the feeder link **510** shown in FIG. **5**. This is before a topology change event such as the ISL deactivation.
(120) As shown by reference number **930**, the NTN entity **810** may transmit an indication of an ISL deactivation. In some aspects, the UE **820** may obtain the indication through a schedule, a calculation, or another communication. The indication may include the ISL deactivation time **902**. The indication may include the transition end time **904**. Alternatively, the UE **820** may calculate the

transition end time based at least in part on an indication or a calculated estimate of the latency change **906**.

(121) As shown by reference number **935**, NTN entity **810** may perform the ISL deactivation (shown by ISL deactivation **936**) at the ISL deactivation time **902**. As shown by reference number **940**, the UE **820** may buffer or may have buffered communications for transmission. The communications may be received from the upper protocol layer **704**. The UE **820** may buffer the communications in a buffer (e.g., buffer **706**) at the upper protocol layer **704** or at the AS layer **702**. As shown by reference number **945**, the UE **820** may release the communications at a rate of release **912** that changes between the ISL deactivation time **902** to the transition end time **904**. The rate of release **912** may be increasing and thus the latency is decreasing. The buffer may release the communications to a lower protocol layer (e.g., MAC layer and/or PHY layer) for transmission. The UE **820** may transmit the communications as they are released from the buffer. In some aspects, the rate of release **912** of the communications from the buffer may be based on a function of time, where a time instance is an input, and a latency and/or the rate of release **912** is the output. The function may be linear function that gradually decreases at a steady rate. The function may be a stepped function where the latency increases a specified amount at instances of time. The function may be a non-linear function or any other function for changing the rate of release **912** between the ISL deactivation time **902** and the transition end time **904**. The UE **820** may change the rate of release by changing a timer of the buffer. By the transition end time **904**, the gradual decrease in latency may have been completed. As a result, there is no sudden change in latency and communications are not disrupted. This reduces latency and conserves processing resources and signaling resources.

(122) As shown by reference number **950**, the UE **820** and the NTN entity **810** may communicate via the service link **430**. As shown by reference number **955**, the NTN entity **810** and the ground station **830** may communicate via feeder link **510**.

(123) While example **800** shows the latency decreasing based on the rate of release **912** because the first propagation delay **908** before ISL activation is greater than the second propagation delay **910** after ISL deactivation, in some aspects, the latency may be increasing if the first propagation delay **908** before ISL deactivation is less than the second propagation delay **910** after ISL deactivation.

(124) In some aspects, the UE **820** may determine to perform a gradual increase or decrease in latency based at least in part on a change in the propagation delay, the type of service, characteristics of the service, a QoS flow, or any combination thereof. For example, if the latency change **806** or **906** satisfies a threshold (e.g., minimum latency change, amount of time), the UE **820** may use the gradual increase or decrease in latency. If the latency change **806** or **906** does not satisfy the threshold, the UE **820** may not use the gradual increase or decrease in latency. The determination may be based on a relative change in the propagation delay, such as the latency change **806** or **906** in proportion to (e.g., divided by) the first propagation delay **808** or **908**. In some aspects, the UE **820** may trigger the gradual increase or decrease in latency based at least in part on a service or QoS being sensitive to a sudden latency change.

(125) In some aspects, the indication of the ISL activation or ISL deactivation may include other information about the transition, such as a duration of a transition phase. The duration may be larger or smaller based at least in part on the size of the latency change **806** or **906**. The indication may include parameters to be used during the transition phase. For example, the parameters may include a function to be used by the UE **820** to gradually adjust the latency during the transition phase. The parameters may include how the UE **820** should use a current buffer at a specific layers, such as at an RLC or a PDCP layer. The parameters may identify the impacted QoS flow(s) or service(s) such that only the UE(s) with an impacted QoS flow or service applies the gradual increase or decrease in latency.

(126) In some aspects, the gradual latency operations described for the UE **820** may be performed by a network entity, such as a network entity that neighbors or is nearby the UE **820**. In some

aspects, the network entity may be one or more of NTN entity **810**, NTN entity **840**, and ground station **830**. In some aspects, the gradual latency operations performed by UE **820** and the gradual latency operations performed by the network entity may be applied for the transmission in different directions. For example, when an ISL is to be activated, the UE **820** may release the received downlink data with an increased latency, while the network entity may release the received uplink data with an increased latency.

(127) As indicated above, FIG. **9** is provided as an example. Other examples may differ from what is described with regard to FIG. **9**.

(128) FIG. **10** is a diagram illustrating an example process **1000** performed, for example, by a wireless communication device, in accordance with the present disclosure. Example process **1000** is an example where the wireless communication device (e.g., UE **820**, a network entity) performs operations associated with buffer release of received communications in association with an ISL transition.

(129) As shown in FIG. **10**, in some aspects, process **1000** may include obtaining, at an AS layer, an indication of an ISL activation that is to take place at an activation time (block **1010**). For example, the wireless communication device (e.g., using communication manager **1208** or **1508** and/or transition component **1210** or **1510** depicted in FIG. **12** or **15**) may obtain, at an AS layer, an indication of an ISL activation that is to take place at an activation time, as described above.

(130) As further shown in FIG. **10**, in some aspects, process **1000** may include receiving communications (block **1020**). For example, the wireless communication device (e.g., using communication manager **1208** or **1508** and/or reception component **1202** or **1502** depicted in FIG. **12** or **15**) may receive communications, as described above.

(131) As further shown in FIG. **10**, in some aspects, process **1000** may include buffering the communications in a buffer at the AS layer (block **1030**). For example, the wireless communication device (e.g., using communication manager **1208** or **1508** and/or buffer component **1212** or **1512** depicted in FIG. **12** or **15**) may buffer the communications in a buffer at the AS layer, as described above.

(132) As further shown in FIG. **10**, in some aspects, process **1000** may include releasing the communications from the buffer to an upper protocol layer at a rate of release that changes between a transition start time and the activation time (block **1040**). For example, the wireless communication device (e.g., using communication manager **1208** or **1508** and/or buffer component **1212** or **1512** depicted in FIG. **12** or **15**) may release the communications from the buffer to an upper protocol layer at a rate of release that changes between a transition start time and the activation time, as described above.

(133) Process **1000** may include additional aspects, such as any single aspect or any combination of aspects described below and/or in connection with one or more other processes described elsewhere herein.

(134) In a first aspect, obtaining the indication includes receiving the indication.

(135) In a second aspect, alone or in combination with the first aspect, process **1000** includes determining the transition start time based at least in part on the indication.

(136) In a third aspect, alone or in combination with one or more of the first and second aspects, process **1000** includes determining, based at least in part on the indication, a difference between a first propagation latency before the ISL activation and a second propagation latency after the ISL activation.

(137) In a fourth aspect, alone or in combination with one or more of the first through third aspects, process **1000** includes determining the transition start time based at least in part on the difference.

(138) In a fifth aspect, alone or in combination with one or more of the first through fourth aspects, a rate of buffering or the rate of release changes based at least in part on the difference satisfying a threshold.

(139) In a sixth aspect, alone or in combination with one or more of the first through fifth aspects,

the rate of release increases in response to the second propagation latency being greater than the first propagation latency.
(140) In a seventh aspect, alone or in combination with one or more of the first through sixth aspects, the rate of release decreases in response to the second propagation latency being less than the first propagation latency.
(141) In an eighth aspect, alone or in combination with one or more of the first through seventh aspects, a rate of buffering or the rate of release changes based at least in part on a configured function.
(142) In a ninth aspect, alone or in combination with one or more of the first through eighth aspects, the configured function is a linear function.
(143) In a tenth aspect, alone or in combination with one or more of the first through ninth aspects, the configured function is a non-linear function.
(144) In an eleventh aspect, alone or in combination with one or more of the first through tenth aspects, the configured function is a stepped function.
(145) In a twelfth aspect, alone or in combination with one or more of the first through eleventh aspects, process **1000** includes changing the rate of release by changing a timer associated with the buffer.
(146) In a thirteenth aspect, alone or in combination with one or more of the first through twelfth aspects, a rate of buffering or the rate of release changes based at least in part on a latency sensitivity of a service or quality of service flow.
(147) In a fourteenth aspect, alone or in combination with one or more of the first through thirteenth aspects, buffering the communications includes buffering the communications using a buffer that is activated based at least in part on receiving the indication of the ISL activation.
(148) In a fifteenth aspect, alone or in combination with one or more of the first through fourteenth aspects, the wireless communication device is a UE.
(149) In a sixteenth aspect, alone or in combination with one or more of the first through fifteenth aspects, the wireless communication device is a network entity, such as a network entity that neighbors a UE.
(150) Although FIG. **10** shows example blocks of process **1000**, in some aspects, process **1000** may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. **10**. Additionally, or alternatively, two or more of the blocks of process **1000** may be performed in parallel.
(151) FIG. **11** is a diagram illustrating an example process **1100** performed, for example, by a UE, in accordance with the present disclosure. Example process **1100** is an example where the UE (e.g., UE **120**, UE **820**) performs operations associated with buffer release of communications for transmission in association with an ISL transition.
(152) As shown in FIG. **11**, in some aspects, process **1100** may include obtaining, at an AS layer, an indication of an ISL deactivation that is to take place at a deactivation time (block **1110**). For example, the UE (e.g., using communication manager **1208** and/or transition component **1210** depicted in FIG. **12**) may obtain, at an AS layer, an indication of an ISL deactivation that is to take place at a deactivation time, as described above.
(153) As further shown in FIG. **11**, in some aspects, process **1100** may include buffering communications in a buffer (block **1120**). For example, the UE (e.g., using communication manager **1208** and/or buffer component **1212** depicted in FIG. **12**) may buffer communications in a buffer, as described above.
(154) As further shown in FIG. **11**, in some aspects, process **1100** may include releasing the communications from the buffer at a rate of release that changes between the deactivation time and a transition end time (block **1130**). For example, the UE (e.g., using communication manager **1208** and/or buffer component **1212** depicted in FIG. **12**) may release the communications from the buffer at a rate of release that changes between the deactivation time and a transition end time, as

described above.
(155) As further shown in FIG. **11**, in some aspects, process **1100** may include transmitting the communications (block **1140**). For example, the UE (e.g., using communication manager **1208** and/or transmission component **1204** depicted in FIG. **12**) may transmit the communications, as described above.
(156) Process **1100** may include additional aspects, such as any single aspect or any combination of aspects described below and/or in connection with one or more other processes described elsewhere herein.
(157) In a first aspect, process **1100** includes determining the transition end time based at least in part on the indication.
(158) In a second aspect, alone or in combination with the first aspect, process **1100** includes determining, based at least in part on the indication, a difference between a first propagation latency before the ISL deactivation and a second propagation latency after the ISL deactivation.
(159) In a third aspect, alone or in combination with one or more of the first and second aspects, process **1100** includes determining the transition end time based at least in part on the difference.
(160) In a fourth aspect, alone or in combination with one or more of the first through third aspects, a rate of buffering or the rate of release changes based at least in part on the difference satisfying a threshold.
(161) In a fifth aspect, alone or in combination with one or more of the first through fourth aspects, a rate of buffering or the rate of release changes based at least in part on a configured function.
(162) In a sixth aspect, alone or in combination with one or more of the first through fifth aspects, process **1100** includes changing the rate of release by changing a timer associated with the buffer.
(163) In a seventh aspect, alone or in combination with one or more of the first through sixth aspects, a rate of buffering or the rate of release changes based at least in part on a latency sensitivity of a service or quality of service flow.
(164) Although FIG. **11** shows example blocks of process **1100**, in some aspects, process **1100** may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. **11**. Additionally, or alternatively, two or more of the blocks of process **1100** may be performed in parallel.
(165) FIG. **12** is a diagram of an example apparatus **1200** for wireless communication, in accordance with the present disclosure. The apparatus **1200** may be a UE (e.g., UE **120**, UE **820**), or a UE may include the apparatus **1200**. In some aspects, the apparatus **1200** includes a reception component **1202** and a transmission component **1204**, which may be in communication with one another (for example, via one or more buses and/or one or more other components). As shown, the apparatus **1200** may communicate with another apparatus **1206** (such as a UE, a base station, or another wireless communication device) using the reception component **1202** and the transmission component **1204**. As further shown, the apparatus **1200** may include the communication manager **1208**. The communication manager **1208** may control and/or otherwise manage one or more operations of the reception component **1202** and/or the transmission component **1204**. In some aspects, the communication manager **1208** may include one or more antennas, a modem, a controller/processor, a memory, or a combination thereof, of the UE described in connection with FIG. **2**. The communication manager **1208** may be, or be similar to, the communication manager **140** depicted in FIGS. **1** and **2**. For example, in some aspects, the communication manager **1208** may be configured to perform one or more of the functions described as being performed by the communication manager **140**. In some aspects, the communication manager **1208** may include the reception component **1202** and/or the transmission component **1204**. The communication manager **1208** may include a transition component **1210** and/or a buffer component **1212**, among other examples.
(166) In some aspects, the apparatus **1200** may be configured to perform one or more operations described herein in connection with FIGS. **1-9**. Additionally, or alternatively, the apparatus **1200**

may be configured to perform one or more processes described herein, such as process **1000** of FIG. **10**, process **1100** of FIG. **11**, or a combination thereof. In some aspects, the apparatus **1200** and/or one or more components shown in FIG. **12** may include one or more components of the UE described in connection with FIG. **2**. Additionally, or alternatively, one or more components shown in FIG. **12** may be implemented within one or more components described in connection with FIG. **2**. Additionally, or alternatively, one or more components of the set of components may be implemented at least in part as software stored in a memory. For example, a component (or a portion of a component) may be implemented as instructions or code stored in a non-transitory computer-readable medium and executable by a controller or a processor to perform the functions or operations of the component.

(167) The reception component **1202** may receive communications, such as reference signals, control information, data communications, or a combination thereof, from the apparatus **1206**. The reception component **1202** may provide received communications to one or more other components of the apparatus **1200**. In some aspects, the reception component **1202** may perform signal processing on the received communications (such as filtering, amplification, demodulation, analog-to-digital conversion, demultiplexing, deinterleaving, de-mapping, equalization, interference cancellation, or decoding, among other examples), and may provide the processed signals to the one or more other components of the apparatus **1200**. In some aspects, the reception component **1202** may include one or more antennas, a modem, a demodulator, a MIMO detector, a receive processor, a controller/processor, a memory, or a combination thereof, of the UE described in connection with FIG. **2**.

(168) The transmission component **1204** may transmit communications, such as reference signals, control information, data communications, or a combination thereof, to the apparatus **1206**. In some aspects, one or more other components of the apparatus **1200** may generate communications and may provide the generated communications to the transmission component **1204** for transmission to the apparatus **1206**. In some aspects, the transmission component **1204** may perform signal processing on the generated communications (such as filtering, amplification, modulation, digital-to-analog conversion, multiplexing, interleaving, mapping, or encoding, among other examples), and may transmit the processed signals to the apparatus **1206**. In some aspects, the transmission component **1204** may include one or more antennas, a modem, a modulator, a transmit MIMO processor, a transmit processor, a controller/processor, a memory, or a combination thereof, of the UE described in connection with FIG. **2**. In some aspects, the transmission component **1204** may be co-located with the reception component **1202** in a transceiver.

(169) In some aspects, the transition component **1210** may obtain, at an AS layer, an indication of an ISL activation that is to take place at an activation time. The reception component **1202** may receive communications. The buffer component **1212** may buffer the communications in a buffer at the AS layer. The buffer component **1212** may release the communications from the buffer to an upper protocol layer at a rate of release that is to change between a transition start time and the activation time.

(170) The transition component **1210** may determine the transition start time based at least in part on the indication. The transition component **1210** may determine, based at least in part on the indication, a difference between a first propagation latency before the ISL activation and a second propagation latency after the ISL activation. The transition component **1210** may determine the transition start time based at least in part on the difference. The buffer component **1212** may change the rate of release by changing a timer associated with the buffer.

(171) In some aspects, the transition component **1210** may obtain, at an AS layer, an indication of an ISL deactivation that is to take place at a deactivation time. The buffer component **1212** may buffer communications in a buffer. The buffer component **1212** may release the communications from the buffer at a rate of release that is to change between the deactivation time and a transition end time. The transmission component **1204** may transmit the communications.

(172) The transition component **1210** may determine the transition end time based at least in part on the indication. The transition component **1210** may determine, based at least in part on the indication, a difference between a first propagation latency before the ISL deactivation and a second propagation latency after the ISL deactivation. The transition component **1210** may determine the transition end time based at least in part on the difference. The buffer component **1212** may change the rate of release by changing a timer associated with the buffer.
(173) The number and arrangement of components shown in FIG. **12** are provided as an example. In practice, there may be additional components, fewer components, different components, or differently arranged components than those shown in FIG. **12**. Furthermore, two or more components shown in FIG. **12** may be implemented within a single component, or a single component shown in FIG. **12** may be implemented as multiple, distributed components. Additionally, or alternatively, a set of (one or more) components shown in FIG. **12** may perform one or more functions described as being performed by another set of components shown in FIG. **12**.
(174) FIG. **13** is a diagram illustrating an example **1300** of a hardware implementation for an apparatus **1305** employing a processing system **1310**, in accordance with the present disclosure. The apparatus **1305** may be a UE.
(175) The processing system **1310** may be implemented with a bus architecture, represented generally by the bus **1315**. The bus **1315** may include any number of interconnecting buses and bridges depending on the specific application of the processing system **1310** and the overall design constraints. The bus **1315** links together various circuits including one or more processors and/or hardware components, represented by the processor **1320**, the illustrated components, and the computer-readable medium/memory **1325**. The bus **1315** may also link various other circuits, such as timing sources, peripherals, voltage regulators, and/or power management circuits.
(176) The processing system **1310** may be coupled to a transceiver **1330**. The transceiver **1330** is coupled to one or more antennas **1335**. The transceiver **1330** provides a means for communicating with various other apparatuses over a transmission medium. The transceiver **1330** receives a signal from the one or more antennas **1335**, extracts information from the received signal, and provides the extracted information to the processing system **1310**, specifically the reception component **1202**. In addition, the transceiver **1330** receives information from the processing system **1310**, specifically the transmission component **1204**, and generates a signal to be applied to the one or more antennas **1335** based at least in part on the received information.
(177) The processing system **1310** includes a processor **1320** coupled to a computer-readable medium/memory **1325**. The processor **1320** is responsible for general processing, including the execution of software stored on the computer-readable medium/memory **1325**. The software, when executed by the processor **1320**, causes the processing system **1310** to perform the various functions described herein for any particular apparatus. The computer-readable medium/memory **1325** may also be used for storing data that is manipulated by the processor **1320** when executing software. The processing system further includes at least one of the illustrated components. The components may be software modules running in the processor **1320**, resident/stored in the computer readable medium/memory **1325**, one or more hardware modules coupled to the processor **1320**, or some combination thereof.
(178) In some aspects, the processing system **1310** may be a component of the UE **120** and may include the memory **282** and/or at least one of the TX MIMO processor **266**, the receive (RX) processor **258**, and/or the controller/processor **280**. In some aspects, the apparatus **1305** for wireless communication includes means for obtaining, at an AS layer, an indication of an ISL activation that is to take place at an activation time; means for receiving communications; means for buffering the communications in a buffer at the AS layer; and/or means for releasing the communications from the buffer to an upper protocol layer at a rate of release that is to change between a transition start time and the activation time. In some aspects, the apparatus **1305** for

wireless communication includes means for obtaining, at an AS layer, an indication of an ISL deactivation that is to take place at a deactivation time; means for buffering communications in a buffer; means for releasing the communications from the buffer at a rate of release that is to change between the deactivation time and a transition end time; and/or means for transmitting the communications. The aforementioned means may be one or more of the aforementioned components of the apparatus **1200** and/or the processing system **1310** of the apparatus **1305** configured to perform the functions recited by the aforementioned means. As described elsewhere herein, the processing system **1310** may include the TX MIMO processor **266**, the RX processor **258**, and/or the controller/processor **280**. In one configuration, the aforementioned means may be the TX MIMO processor **266**, the RX processor **258**, and/or the controller/processor **280** configured to perform the functions and/or operations recited herein.

(179) In some aspects, the processing system **1310** may be a component of the base station **110** and may include the memory **242** and/or at least one of the TX MIMO processor **230**, the RX processor **238**, and/or the controller/processor **240**. In some aspects, the apparatus **1305** for wireless communication includes means for obtaining, at an AS layer, an indication of an ISL activation that is to take place at an activation time; means for receiving communications; means for buffering the communications in a buffer at the AS layer; and/or means for releasing the communications from the buffer to an upper protocol layer at a rate of release that is to change between a transition start time and the activation time. In some aspects, the apparatus **1305** for wireless communication includes means for obtaining, at an AS layer, an indication of an ISL deactivation that is to take place at a deactivation time; means for buffering communications in a buffer; means for releasing the communications from the buffer at a rate of release that is to change between the deactivation time and a transition end time; and/or means for transmitting the communications. The aforementioned means may be one or more of the aforementioned components of the apparatus **1200** and/or the processing system **1310** of the apparatus **1305** configured to perform the functions recited by the aforementioned means. As described elsewhere herein, the processing system **1310** may include the TX MIMO processor **230**, the receive processor **238**, and/or the controller/processor **240**. In one configuration, the aforementioned means may be the TX MIMO processor **230**, the receive processor **238**, and/or the controller/processor **240** configured to perform the functions and/or operations recited herein.

(180) FIG. **13** is provided as an example. Other examples may differ from what is described in connection with FIG. **13**.

(181) FIG. **14** is a diagram illustrating an example **1400** of an implementation of code and circuitry for an apparatus **1405**, in accordance with the present disclosure. The apparatus **1405** may be a UE, or a UE may include the apparatus **1405**.

(182) As shown in FIG. **14**, the apparatus **1405** may include circuitry for obtaining, at an AS layer, an indication of an ISL transition (e.g., activation that is to take place at an activation time) (circuitry **1420**). For example, the circuitry **1420** may enable the apparatus **1405** to obtain, at an AS layer, an indication of an ISL activation that is to take place at an activation time.

(183) As shown in FIG. **14**, the apparatus **1405** may include, stored in computer-readable medium **1325**, code for obtaining, at an AS layer, an indication of an ISL transition (e.g., activation that is to take place at an activation time) (code **1425**). For example, the code **1425**, when executed by processor **1320**, may cause processor **1320** to cause transceiver **1330** to obtain, at an AS layer, an indication of an ISL activation that is to take place at an activation time.

(184) As shown in FIG. **14**, the apparatus **1405** may include circuitry for receiving communications (circuitry **1430**). For example, the circuitry **1430** may enable the apparatus **1405** to receive communications.

(185) As shown in FIG. **14**, the apparatus **1405** may include, stored in computer-readable medium **1325**, code for receiving communications (code **1435**). For example, the code **1435**, when executed by processor **1320**, may cause processor **1320** to cause transceiver **1330** to receive

communications.
(186) As shown in FIG. **14**, the apparatus **1405** may include circuitry for buffering the communications in a buffer at the AS layer (circuitry **1440**). For example, the circuitry **1440** may enable the apparatus **1405** to buffer the communications in a buffer at the AS layer.
(187) As shown in FIG. **14**, the apparatus **1405** may include, stored in computer-readable medium **1325**, code for buffering the communications in a buffer at the AS layer (code **1445**). For example, the code **1445**, when executed by processor **1320**, may cause processor **1320** to buffer the communications in a buffer at the AS layer.
(188) As shown in FIG. **14**, the apparatus **1405** may include circuitry for releasing the communications from the buffer to an upper protocol layer at a rate of release that changes between a transition start time and the activation time (circuitry **1450**). For example, the circuitry **1450** may enable the apparatus **1405** to release the communications from the buffer to an upper protocol layer at a rate of release that changes between a transition start time and the activation time.
(189) As shown in FIG. **14**, the apparatus **1405** may include, stored in computer-readable medium **1325**, code for releasing the communications from the buffer to an upper protocol layer at a rate of release that changes between a transition start time and the activation time (code **1455**). For example, the code **1455**, when executed by processor **1320**, may cause processor **1320** to release the communications from the buffer to an upper protocol layer at a rate of release that changes between a transition start time and the activation time.
(190) In some aspects, the apparatus **1405** may include circuitry for obtaining, at an AS layer, an indication of an ISL transition (e.g., deactivation that is to take place at a deactivation time) (circuitry **1420**). For example, the circuitry **1420** may enable the apparatus **1405** to obtain, at an AS layer, an indication of an ISL deactivation that is to take place at a deactivation time.
(191) As shown in FIG. **14**, the apparatus **1405** may include, stored in computer-readable medium **1325**, code for obtaining, at an AS layer, an indication of an ISL transition (e.g., deactivation that is to take place at a deactivation time) (code **1425**). For example, the code **1425**, when executed by processor **1320**, may cause processor **1320** to cause transceiver **1330** to obtain, at an AS layer, an indication of an ISL transition (e.g., deactivation that is to take place at a deactivation time).
(192) As shown in FIG. **14**, the apparatus **1405** may include circuitry for buffering communications in a buffer (circuitry **1440**). For example, the circuitry **1430** may enable the apparatus **1405** to buffer communications in a buffer.
(193) As shown in FIG. **14**, the apparatus **1405** may include, stored in computer-readable medium **1325**, code for buffering communications in a buffer (code **1445**). For example, the code **1435**, when executed by processor **1320**, may cause processor **1320** to buffer communications in a buffer.
(194) As shown in FIG. **14**, the apparatus **1405** may include circuitry for releasing the communications from the buffer at a rate of release that changes between the deactivation time and a transition end time (circuitry **1450**). For example, the circuitry **1440** may enable the apparatus **1405** to release the communications from the buffer at a rate of release that changes between the deactivation time and a transition end time.
(195) As shown in FIG. **14**, the apparatus **1405** may include, stored in computer-readable medium **1325**, code for releasing the communications from the buffer at a rate of release that changes between the deactivation time and a transition end time (code **1455**). For example, the code **1445**, when executed by processor **1320**, may cause processor **1320** to release the communications from the buffer at a rate of release that changes between the deactivation time and a transition end time.
(196) As shown in FIG. **14**, the apparatus **1405** may include circuitry for transmitting the communications (circuitry **1460**). For example, the circuitry **1450** may enable the apparatus **1405** to transmit the communications.
(197) As shown in FIG. **14**, the apparatus **1405** may include, stored in computer-readable medium **1325**, code for transmitting the communications (code **1465**). For example, the code **1455**, when executed by processor **1320**, may cause processor **1320** to cause transceiver **1330** to transmit the

communications.

(198) FIG. **14** is provided as an example. Other examples may differ from what is described in connection with FIG. **14**.

(199) FIG. **15** is a diagram of an example apparatus **1500** for wireless communication, in accordance with the present disclosure. The apparatus **1500** may be a network entity (e.g., base station **110**, gateway **450**, satellite **440**, satellite **490**), or a network entity may include the apparatus **1500**. In some aspects, the apparatus **1500** includes a reception component **1502** and a transmission component **1504**, which may be in communication with one another (for example, via one or more buses and/or one or more other components). As shown, the apparatus **1500** may communicate with another apparatus **1506** (such as a UE, a base station, or another wireless communication device) using the reception component **1502** and the transmission component **1504**. As further shown, the apparatus **1500** may include the communication manager **1508**. The communication manager **1508** may control and/or otherwise manage one or more operations of the reception component **1502** and/or the transmission component **1504**. In some aspects, the communication manager **1508** may include one or more antennas, a modem, a controller/processor, a memory, or a combination thereof, of the network entity described in connection with FIG. **2**. The communication manager **1508** may be, or be similar to, the communication manager **150** depicted in FIGS. **1** and **2**. For example, in some aspects, the communication manager **1508** may be configured to perform one or more of the functions described as being performed by the communication manager **150**. In some aspects, the communication manager **1508** may include the reception component **1502** and/or the transmission component **1504**. The communication manager **1508** may include a transition component **1510** and/or a buffer component **1512**, among other examples.

(200) In some aspects, the apparatus **1500** may be configured to perform one or more operations described herein in connection with FIGS. **1-9**. Additionally, or alternatively, the apparatus **1500** may be configured to perform one or more processes described herein, such as process **1000** of FIG. **10**, process **1100** of FIG. **11**, or a combination thereof. In some aspects, the apparatus **1500** and/or one or more components shown in FIG. **15** may include one or more components of the network entity described in connection with FIG. **2**. Additionally, or alternatively, one or more components shown in FIG. **15** may be implemented within one or more components described in connection with FIG. **2**. Additionally, or alternatively, one or more components of the set of components may be implemented at least in part as software stored in a memory. For example, a component (or a portion of a component) may be implemented as instructions or code stored in a non-transitory computer-readable medium and executable by a controller or a processor to perform the functions or operations of the component.

(201) The reception component **1502** may receive communications, such as reference signals, control information, data communications, or a combination thereof, from the apparatus **1506**. The reception component **1502** may provide received communications to one or more other components of the apparatus **1500**. In some aspects, the reception component **1502** may perform signal processing on the received communications (such as filtering, amplification, demodulation, analog-to-digital conversion, demultiplexing, deinterleaving, de-mapping, equalization, interference cancellation, or decoding, among other examples), and may provide the processed signals to the one or more other components of the apparatus **1500**. In some aspects, the reception component **1502** may include one or more antennas, a modem, a demodulator, a MIMO detector, a receive processor, a controller/processor, a memory, or a combination thereof, of the network entity described in connection with FIG. **2**.

(202) The transmission component **1504** may transmit communications, such as reference signals, control information, data communications, or a combination thereof, to the apparatus **1506**. In some aspects, one or more other components of the apparatus **1500** may generate communications and may provide the generated communications to the transmission component **1504** for

transmission to the apparatus **1506**. In some aspects, the transmission component **1504** may perform signal processing on the generated communications (such as filtering, amplification, modulation, digital-to-analog conversion, multiplexing, interleaving, mapping, or encoding, among other examples), and may transmit the processed signals to the apparatus **1506**. In some aspects, the transmission component **1504** may include one or more antennas, a modem, a modulator, a transmit MIMO processor, a transmit processor, a controller/processor, a memory, or a combination thereof, of the network entity described in connection with FIG. **2**. In some aspects, the transmission component **1504** may be co-located with the reception component **1502** in a transceiver.
(203) In some aspects, the transition component **1510** may obtain, at an AS layer, an indication of an ISL activation that is to take place at an activation time. The reception component **1502** may receive communications. The buffer component **1512** may buffer the communications in a buffer at the AS layer. The buffer component **1512** may release the communications from the buffer to an upper protocol layer at a rate of release that changes between a transition start time and the activation time.
(204) The transition component **1510** may determine the transition start time based at least in part on the indication. The transition component **1510** may determine, based at least in part on the indication, a difference between a first propagation latency before the ISL activation and a second propagation latency after the ISL activation. The transition component **1510** may determine the transition start time based at least in part on the difference. The buffer component **1512** may change the rate of release by changing a timer associated with the buffer.
(205) In some aspects, the transition component **1510** may obtain, at an AS layer, an indication of an ISL deactivation that is to take place at a deactivation time. The buffer component **1512** may buffer communications in a buffer. The buffer component **1512** may release the communications from the buffer at a rate of release that changes between the deactivation time and a transition end time. The transmission component **1504** may transmit the communications.
(206) The transition component **1510** may determine the transition end time based at least in part on the indication. The transition component **1510** may determine, based at least in part on the indication, a difference between a first propagation latency before the ISL deactivation and a second propagation latency after the ISL deactivation.
(207) The transition component **1510** may determine the transition end time based at least in part on the difference. The buffer component **1512** may change the rate of release by changing a timer associated with the buffer.
(208) The number and arrangement of components shown in FIG. **15** are provided as an example. In practice, there may be additional components, fewer components, different components, or differently arranged components than those shown in FIG. **15**. Furthermore, two or more components shown in FIG. **15** may be implemented within a single component, or a single component shown in FIG. **15** may be implemented as multiple, distributed components. Additionally, or alternatively, a set of (one or more) components shown in FIG. **15** may perform one or more functions described as being performed by another set of components shown in FIG. **15**.
(209) FIG. **16** is a diagram illustrating an example **1600** of a hardware implementation for an apparatus **1605** employing a processing system **1610**, in accordance with the present disclosure. The apparatus **1605** may be a network entity.
(210) The processing system **1610** may be implemented with a bus architecture, represented generally by the bus **1615**. The bus **1615** may include any number of interconnecting buses and bridges depending on the specific application of the processing system **1610** and the overall design constraints. The bus **1615** links together various circuits including one or more processors and/or hardware components, represented by the processor **1620**, the illustrated components, and the computer-readable medium/memory **1625**. The bus **1615** may also link various other circuits, such as timing sources, peripherals, voltage regulators, and/or power management circuits.

(211) The processing system **1610** may be coupled to a transceiver **1630**. The transceiver **1630** is coupled to one or more antennas **1635**. The transceiver **1630** provides a means for communicating with various other apparatuses over a transmission medium. The transceiver **1630** receives a signal from the one or more antennas **1635**, extracts information from the received signal, and provides the extracted information to the processing system **1610**, specifically the reception component **1502**. In addition, the transceiver **1630** receives information from the processing system **1610**, specifically the transmission component **1504**, and generates a signal to be applied to the one or more antennas **1635** based at least in part on the received information.
(212) The processing system **1610** includes a processor **1620** coupled to a computer-readable medium/memory **1625**. The processor **1620** is responsible for general processing, including the execution of software stored on the computer-readable medium/memory **1625**. The software, when executed by the processor **1620**, causes the processing system **1610** to perform the various functions described herein for any particular apparatus. The computer-readable medium/memory **1625** may also be used for storing data that is manipulated by the processor **1620** when executing software. The processing system further includes at least one of the illustrated components. The components may be software modules running in the processor **1620**, resident/stored in the computer readable medium/memory **1625**, one or more hardware modules coupled to the processor **1620**, or some combination thereof.
(213) In some aspects, the processing system **1610** may be a component of the UE **120** and may include the memory **282** and/or at least one of the TX MIMO processor **266**, the RX processor **258**, and/or the controller/processor **280**. In some aspects, the apparatus **1605** for wireless communication includes means for obtaining, at an AS layer, an indication of an ISL activation that is to take place at an activation time; means for receiving communications; means for buffering the communications in a buffer at the AS layer; and/or means for releasing the communications from the buffer to an upper protocol layer at a rate of release that changes between a transition start time and the activation time. In some aspects, the apparatus **1605** for wireless communication includes means for obtaining, at an AS layer, an indication of an ISL deactivation that is to take place at a deactivation time; means for buffering communications in a buffer; means for releasing the communications from the buffer at a rate of release that changes between the deactivation time and a transition end time; and/or means for transmitting the communications. The aforementioned means may be one or more of the aforementioned components of the apparatus **1500** and/or the processing system **1610** of the apparatus **1605** configured to perform the functions recited by the aforementioned means. As described elsewhere herein, the processing system **1610** may include the TX MIMO processor **266**, the RX processor **258**, and/or the controller/processor **280**. In one configuration, the aforementioned means may be the TX MIMO processor **266**, the RX processor **258**, and/or the controller/processor **280** configured to perform the functions and/or operations recited herein.
(214) In some aspects, the processing system **1610** may be a component of the base station **110** and may include the memory **242** and/or at least one of the TX MIMO processor **230**, the RX processor **238**, and/or the controller/processor **240**. In some aspects, the apparatus **1605** for wireless communication includes means for obtaining, at an AS layer, an indication of an ISL activation that is to take place at an activation time; means for receiving communications; means for buffering the communications in a buffer at the AS layer; and/or means for releasing the communications from the buffer to an upper protocol layer at a rate of release that changes between a transition start time and the activation time. In some aspects, the apparatus **1605** for wireless communication includes means for obtaining, at an AS layer, an indication of an ISL deactivation that is to take place at a deactivation time; means for buffering communications in a buffer; means for releasing the communications from the buffer at a rate of release that changes between the deactivation time and a transition end time; and/or means for transmitting the communications. The aforementioned means may be one or more of the aforementioned components of the apparatus **1500** and/or the

processing system **1610** of the apparatus **1605** configured to perform the functions recited by the aforementioned means. As described elsewhere herein, the processing system **1610** may include the TX MIMO processor **230**, the receive processor **238**, and/or the controller/processor **240**. In one configuration, the aforementioned means may be the TX MIMO processor **230**, the receive processor **238**, and/or the controller/processor **240** configured to perform the functions and/or operations recited herein.
(215) FIG. **16** is provided as an example. Other examples may differ from what is described in connection with FIG. **16**.
(216) FIG. **17** is a diagram illustrating an example **1700** of an implementation of code and circuitry for an apparatus **1705**, in accordance with the present disclosure. The apparatus **1705** may be a network entity, or a network entity may include the apparatus **1705**.
(217) As shown in FIG. **17**, the apparatus **1705** may include circuitry for obtaining, at an AS layer, an indication of an ISL transition (e.g., activation that is to take place at an activation time) (circuitry **1720**). For example, the circuitry **1720** may enable the apparatus **1705** to obtain, at an AS layer, an indication of an ISL activation that is to take place at an activation time.
(218) As shown in FIG. **17**, the apparatus **1705** may include, stored in computer-readable medium **1625**, code for obtaining, at an AS layer, an indication of an ISL transition (e.g., activation that is to take place at an activation time) (code **1725**). For example, the code **1725**, when executed by processor **1620**, may cause processor **1620** to cause transceiver **1630** to obtain, at an AS layer, an indication of an ISL activation that is to take place at an activation time.
(219) As shown in FIG. **17**, the apparatus **1705** may include circuitry for receiving communications (circuitry **1730**). For example, the circuitry **1730** may enable the apparatus **1705** to receive communications.
(220) As shown in FIG. **17**, the apparatus **1705** may include, stored in computer-readable medium **1625**, code for receiving communications (code **1735**). For example, the code **1735**, when executed by processor **1620**, may cause processor **1620** to cause transceiver **1630** to receive communications.
(221) As shown in FIG. **17**, the apparatus **1705** may include circuitry for buffering the communications in a buffer at the AS layer (circuitry **1740**). For example, the circuitry **1740** may enable the apparatus **1705** to buffer the communications in a buffer at the AS layer.
(222) As shown in FIG. **17**, the apparatus **1705** may include, stored in computer-readable medium **1625**, code for buffering the communications in a buffer at the AS layer (code **1745**). For example, the code **1745**, when executed by processor **1620**, may cause processor **1620** to buffer the communications in a buffer at the AS layer.
(223) As shown in FIG. **17**, the apparatus **1705** may include circuitry for releasing the communications from the buffer to an upper protocol layer at a rate of release that changes between a transition start time and the activation time (circuitry **1750**). For example, the circuitry **1750** may enable the apparatus **1705** to release the communications from the buffer to an upper protocol layer at a rate of release that changes between a transition start time and the activation time.
(224) As shown in FIG. **17**, the apparatus **1705** may include, stored in computer-readable medium **1625**, code for releasing the communications from the buffer to an upper protocol layer at a rate of release that changes between a transition start time and the activation time (code **1755**). For example, the code **1755**, when executed by processor **1620**, may cause processor **1620** to release the communications from the buffer to an upper protocol layer at a rate of release that changes between a transition start time and the activation time.
(225) In some aspects, the apparatus **1705** may be a network entity, or a network entity may include the apparatus **1705**.
(226) As shown in FIG. **17**, the apparatus **1705** may include circuitry for obtaining, at an AS layer, an indication of an ISL transition (e.g., deactivation that is to take place at a deactivation time) (circuitry **1720**). For example, the circuitry **1720** may enable the apparatus **1705** to obtain, at an AS

layer, an indication of an ISL deactivation that is to take place at a deactivation time.
(227) As shown in FIG. **17**, the apparatus **1705** may include, stored in computer-readable medium **1625**, code for obtaining, at an AS layer, an indication of an ISL transition (e.g., deactivation that is to take place at a deactivation time) (code **1725**). For example, the code **1725**, when executed by processor **1620**, may cause processor **1620** to cause transceiver **1630** to obtain, at an AS layer, an indication of an ISL transition (e.g., deactivation that is to take place at a deactivation time).
(228) As shown in FIG. **17**, the apparatus **1705** may include circuitry for buffering communications in a buffer (circuitry **1740**). For example, the circuitry **1730** may enable the apparatus **1705** to buffer communications in a buffer.
(229) As shown in FIG. **17**, the apparatus **1705** may include, stored in computer-readable medium **1625**, code for buffering communications in a buffer (code **1745**). For example, the code **1735**, when executed by processor **1620**, may cause processor **1620** to buffer communications in a buffer.
(230) As shown in FIG. **17**, the apparatus **1705** may include circuitry for releasing the communications from the buffer at a rate of release that changes between the deactivation time and a transition end time (circuitry **1750**). For example, the circuitry **1740** may enable the apparatus **1705** to release the communications from the buffer at a rate of release that changes between the deactivation time and a transition end time.
(231) As shown in FIG. **17**, the apparatus **1705** may include, stored in computer-readable medium **1625**, code for releasing the communications from the buffer at a rate of release that changes between the deactivation time and a transition end time (code **1755**). For example, the code **1745**, when executed by processor **1620**, may cause processor **1620** to release the communications from the buffer at a rate of release that changes between the deactivation time and a transition end time.
(232) As shown in FIG. **17**, the apparatus **1705** may include circuitry for transmitting the communications (circuitry **1760**). For example, the circuitry **1750** may enable the apparatus **1705** to transmit the communications.
(233) As shown in FIG. **17**, the apparatus **1705** may include, stored in computer-readable medium **1625**, code for transmitting the communications (code **1765**). For example, the code **1755**, when executed by processor **1620**, may cause processor **1620** to cause transceiver **1630** to transmit the communications.
(234) FIG. **17** is provided as an example. Other examples may differ from what is described in connection with FIG. **17**.
(235) The following provides an overview of some Aspects of the present disclosure:
(236) Aspect 1: A method of wireless communication performed by a wireless communication device, comprising: obtaining, at an access stratum (AS) layer, an indication of an inter-satellite link (ISL) activation that is to take place at an activation time; receiving communications; buffering the communications in a buffer at the AS layer; and releasing the communications from the buffer to an upper protocol layer at a rate of release that changes between a transition start time and the activation time.
(237) Aspect 2: The method of Aspect 1, wherein obtaining the indication includes receiving the indication.
(238) Aspect 3: The method of Aspect 1 or 2, further comprising determining the transition start time based at least in part on the indication.
(239) Aspect 4: The method of any of Aspects 1-3, further comprising determining, based at least in part on the indication, a difference between a first propagation latency before the ISL activation and a second propagation latency after the ISL activation.
(240) Aspect 5: The method of Aspect 4, further comprising determining the transition start time based at least in part on the difference.
(241) Aspect 6: The method of Aspect 4 or 5, wherein a rate of buffering or the rate of release changes based at least in part on the difference satisfying a threshold.
(242) Aspect 7: The method of any of Aspects 4-6, wherein the rate of release increases in response

to the second propagation latency being greater than the first propagation latency.
(243) Aspect 8: The method of any of Aspects 4-6, wherein the rate of release decreases in response to the second propagation latency being less than the first propagation latency.
(244) Aspect 9: The method of any of Aspects 1-8, wherein a rate of buffering or the rate of release changes based at least in part on a configured function.
(245) Aspect 10: The method of Aspect 9, wherein the configured function is a linear function.
(246) Aspect 11: The method of Aspect 9, wherein the configured function is a non-linear function.
(247) Aspect 12: The method of Aspect 9, wherein the configured function is a stepped function.
(248) Aspect 13: The method of any of Aspects 1-12, further comprising changing the rate of release by changing a timer associated with the buffer.
(249) Aspect 14: The method of any of Aspects 1-13, wherein a rate of buffering or the rate of release changes based at least in part on a latency sensitivity of a service or quality of service flow.
(250) Aspect 15: The method of any of Aspects 1-4, wherein buffering the communications includes buffering the communications using a buffer that is activated based at least in part on receiving the indication of the ISL activation.
(251) Aspect 16: The method of any of Aspects 1-15, wherein the wireless communication device is a user equipment.
(252) Aspect 17: The method of any of Aspects 1-15, wherein the wireless communication device is a network entity.
(253) Aspect 18: A method of wireless communication performed by a user equipment (UE), comprising: obtaining, at an access stratum (AS) layer, an indication of an inter-satellite link (ISL) deactivation that is to take place at a deactivation time; buffering communications in a buffer; releasing the communications from the buffer at a rate of release that changes between the deactivation time and a transition end time; and transmitting the communications.
(254) Aspect 19: The method of Aspect 18, further comprising determining the transition end time based at least in part on the indication.
(255) Aspect 20: The method of Aspect 18 or 19, further comprising determining, based at least in part on the indication, a difference between a first propagation latency before the ISL deactivation and a second propagation latency after the ISL deactivation.
(256) Aspect 21: The method of Aspect 20, further comprising determining the transition end time based at least in part on the difference.
(257) Aspect 22: The method of Aspect 20 or 21, wherein a rate of buffering or the rate of release changes based at least in part on the difference satisfying a threshold.
(258) Aspect 23: The method of any of Aspects 20-22, wherein a rate of buffering or the rate of release changes based at least in part on a configured function.
(259) Aspect 24: The method of any of Aspects 20-23, further comprising changing the rate of release by changing a timer associated with the buffer.
(260) Aspect 25: The method of any of Aspects 18-24, wherein a rate of buffering or the rate of release changes based at least in part on a latency sensitivity of a service or quality of service flow.
(261) Aspect 26: An apparatus for wireless communication at a device, comprising a processor; memory coupled with the processor; and instructions stored in the memory and executable by the processor to cause the apparatus to perform the method of one or more of Aspects 1-25.
(262) Aspect 27: A device for wireless communication, comprising a memory and one or more processors coupled to the memory, the one or more processors configured to perform the method of one or more of Aspects 1-25.
(263) Aspect 28: An apparatus for wireless communication, comprising at least one means for performing the method of one or more of Aspects 1-25.
(264) Aspect 29: A non-transitory computer-readable medium storing code for wireless communication, the code comprising instructions executable by a processor to perform the method of one or more of Aspects 1-25.

(265) Aspect 30: A non-transitory computer-readable medium storing a set of instructions for wireless communication, the set of instructions comprising one or more instructions that, when executed by one or more processors of a device, cause the device to perform the method of one or more of Aspects 1-25.

(266) The foregoing disclosure provides illustration and description but is not intended to be exhaustive or to limit the aspects to the precise forms disclosed. Modifications and variations may be made in light of the above disclosure or may be acquired from practice of the aspects.

(267) As used herein, the term “component” is intended to be broadly construed as hardware and/or a combination of hardware and software. “Software” shall be construed broadly to mean instructions, instruction sets, code, code segments, program code, programs, subprograms, software modules, applications, software applications, software packages, routines, subroutines, objects, executables, threads of execution, procedures, and/or functions, among other examples, whether referred to as software, firmware, middleware, microcode, hardware description language, or otherwise. As used herein, a “processor” is implemented in hardware and/or a combination of hardware and software. It will be apparent that systems and/or methods described herein may be implemented in different forms of hardware and/or a combination of hardware and software. The actual specialized control hardware or software code used to implement these systems and/or methods is not limiting of the aspects. Thus, the operation and behavior of the systems and/or methods are described herein without reference to specific software code, since those skilled in the art will understand that software and hardware can be designed to implement the systems and/or methods based, at least in part, on the description herein.

(268) As used herein, “satisfying a threshold” may, depending on the context, refer to a value being greater than the threshold, greater than or equal to the threshold, less than the threshold, less than or equal to the threshold, equal to the threshold, not equal to the threshold, or the like.

(269) Even though particular combinations of features are recited in the claims and/or disclosed in the specification, these combinations are not intended to limit the disclosure of various aspects. Many of these features may be combined in ways not specifically recited in the claims and/or disclosed in the specification. The disclosure of various aspects includes each dependent claim in combination with every other claim in the claim set. As used herein, a phrase referring to “at least one of” a list of items refers to any combination of those items, including single members. As an example, “at least one of: a, b, or c” is intended to cover a, b, c, a+b, a+c, b+c, and a+b+c, as well as any combination with multiples of the same element (e.g., a+a, a+a+a, a+a+b, a+a+c, a+b+b, a+c+c, b+b, b+b+b, b+b+c, c+c, and c+c+c, or any other ordering of a, b, and c).

(270) No element, act, or instruction used herein should be construed as critical or essential unless explicitly described as such. Also, as used herein, the articles “a” and “an” are intended to include one or more items and may be used interchangeably with “one or more.” Further, as used herein, the article “the” is intended to include one or more items referenced in connection with the article “the” and may be used interchangeably with “the one or more.” Furthermore, as used herein, the terms “set” and “group” are intended to include one or more items and may be used interchangeably with “one or more.” Where only one item is intended, the phrase “only one” or similar language is used. Also, as used herein, the terms “has,” “have,” “having,” or the like are intended to be open-ended terms that do not limit an element that they modify (e.g., an element “having” A may also have B). Further, the phrase “based on” is intended to mean “based, at least in part, on” unless explicitly stated otherwise. Also, as used herein, the term “or” is intended to be inclusive when used in a series and may be used interchangeably with “and/or,” unless explicitly stated otherwise (e.g., if used in combination with “either” or “only one of”).

Claims

1. A wireless communication device for wireless communication, comprising: a memory; and one or more processors, coupled to the memory, configured to: obtain, at an access stratum (AS) layer, an indication of an inter-satellite link (ISL) activation that is to take place at an ISL activation time; receive communications; buffer the communications in a buffer at the AS layer; and release the communications from the buffer to an upper protocol layer at a rate of release that is to change between a transition start time and the ISL activation time, wherein a transition that starts at the transition start time is associated with a latency change from a first propagation latency before the ISL activation to a second propagation latency after the ISL activation.
2. The wireless communication device of claim 1, wherein the one or more processors, to obtain the indication, are configured to receive the indication.
3. The wireless communication device of claim 1, wherein the one or more processors are further configured to determine the transition start time based at least in part on the indication.
4. The wireless communication device of claim 1, wherein the one or more processors are further configured to determine, based at least in part on the indication, a difference between the first propagation latency before the ISL activation and the second propagation latency after the ISL activation.
5. The wireless communication device of claim 4, wherein the one or more processors are further configured to determine the transition start time based at least in part on the difference.
6. The wireless communication device of claim 4, wherein a rate of buffering or the rate of release changes based at least in part on whether the difference satisfies a threshold.
7. The wireless communication device of claim 4, wherein the rate of release increases when the second propagation latency is greater than the first propagation latency.
8. The wireless communication device of claim 4, wherein the rate of release decreases when the second propagation latency is less than the first propagation latency.
9. The wireless communication device of claim 1, wherein a rate of buffering or the rate of release changes based at least in part on a configured function.
10. The wireless communication device of claim 9, wherein the configured function is a linear function.
11. The wireless communication device of claim 9, wherein the configured function is a non-linear function.
12. The wireless communication device of claim 9, wherein the configured function is a stepped function.
13. The wireless communication device of claim 1, wherein the one or more processors are further configured to change a time associated with the buffer to change the rate of release.
14. The wireless communication device of claim 1, wherein a rate of buffering or the rate of release changes based at least in part on a latency sensitivity of a service or quality of service flow.
15. The wireless communication device of claim 1, wherein the one or more processors, to buffer the communications, are configured to buffer the communications via a buffer that is activated based at least in part on the indication of the ISL activation.
16. The wireless communication device of claim 1, wherein the wireless communication device is a user equipment.
17. The wireless communication device of claim 1, wherein the wireless communication device is a network entity.
18. A wireless communication device for wireless communication, comprising: a memory; and one or more processors, coupled to the memory, configured to: obtain, at an access stratum (AS) layer, an indication of an inter-satellite link (ISL) deactivation that is to take place at an ISL deactivation time; buffer communications in a buffer; release the communications from the buffer at a rate of release that is to change between the ISL deactivation time and a transition end time, wherein a transition that ends at the transition end time is associated with a latency change from a first

propagation latency before the ISL deactivation to a second propagation latency after the ISL deactivation; and transmit the communications.
19. The wireless communication device of claim 18, wherein the one or more processors are further configured to determine the transition end time based at least in part on the indication.
20. The wireless communication device of claim 18, wherein the one or more processors are further configured to determine, based at least in part on the indication, a difference between the first propagation latency before the ISL deactivation and the second propagation latency after the ISL deactivation.
21. The wireless communication device of claim 20, wherein the one or more processors are further configured to determine the transition end time based at least in part on the difference.
22. The wireless communication device of claim 20, wherein a rate of buffering or the rate of release changes based at least in part on whether the difference satisfies a threshold.
23. The wireless communication device of claim 20, wherein a rate of buffering or the rate of release changes based at least in part on a configured function.
24. The wireless communication device of claim 20, wherein the one or more processors are further configured to change a time associated with the buffer to change the rate of release.
25. The wireless communication device of claim 18, wherein a rate of buffering or the rate of release changes based at least in part on a latency sensitivity of a service or quality of service flow.
26. A method of wireless communication performed at a wireless communication device, comprising: obtaining, at an access stratum (AS) layer, an indication of an inter-satellite link (ISL) activation that is to take place at an ISL activation time; receiving communications; buffering the communications in a buffer at the AS layer; and releasing the communications from the buffer to an upper protocol layer at a rate of release that changes between a transition start time and the ISL activation time, wherein a transition that starts at the transition start time is associated with a latency change from a first propagation latency before the ISL activation to a second propagation latency after the ISL activation.
27. The method of claim 26, further comprising determining the transition start time based at least in part on the indication.
28. The method of claim 26, wherein a rate of buffering or the rate of release changes based at least in part on a configured function.
29. A method of wireless communication performed at a wireless communication device, comprising: obtaining, at an access stratum (AS) layer, an indication of an inter-satellite link (ISL) deactivation that is to take place at an ISL deactivation time; buffering communications in a buffer; releasing the communications from the buffer at a rate of release that changes between the ISL deactivation time and a transition end time, wherein a transition that ends at the transition end time is associated with a latency change from a first propagation latency before the ISL deactivation to a second propagation latency after the ISL deactivation; and transmitting the communications.
30. The method of claim 29, further comprising determining the transition end time based at least in part on the indication.
31. A non-transitory computer-readable medium storing a set of instructions for wireless communication, the set of instructions comprising: one or more instructions that, when executed by one or more processors of a wireless communication device, cause the wireless communication device to: obtain, at an access stratum (AS) layer, an indication of an inter-satellite link (ISL) activation that is to take place at an ISL activation time; receive communications; buffer the communications in a buffer at the AS layer; and release the communications from the buffer to an upper protocol layer at a rate of release that is to change between a transition start time and the ISL activation time, wherein a transition that starts at the transition start time is associated with a latency change from a first propagation latency before the ISL activation to a second propagation latency after the ISL activation.
32. A non-transitory computer-readable medium storing a set of instructions for wireless

communication, the set of instructions comprising: one or more instructions that, when executed by one or more processors of a wireless communication device, cause the wireless communication device to: obtain, at an access stratum (AS) layer, an indication of an inter-satellite link (ISL) deactivation that is to take place at an ISL deactivation time; buffer communications in a buffer; release the communications from the buffer at a rate of release that is to change between the ISL deactivation time and a transition end time, wherein a transition that ends at the transition end time is associated with a latency change from a first propagation latency before the ISL deactivation to a second propagation latency after the ISL deactivation; and transmit the communications.